QRUM Astrophysical Frequency Survey

A Unified Statistical Analysis of Compact Object Frequency Distributions

c 2026 P. Gabor

License:

Creative Commons

Attribution - NonCommercial - NoDerivatives 4.0

International (CC BY -NC -ND 4.0)

Abstract

The astrophysical Universe exhibits an extraordinary diversity of oscillatory phenomena spanning more than twelve orders of magnitude in frequency. White dwarf pulsations, neutron-star spin frequencies, magnetar oscillations, black-hole high-frequency quasi-periodic oscillations (HFQPOs), and other compact-object variability have traditionally been investigated as independent physical phenomena within their respective astrophysical contexts.

The present work explores an alternative perspective.

Rather than analysing each compact-object class separately, we investigate whether a unified statistical framework can reveal common structural organization within their frequency distributions. The objective is not to replace established astrophysical models, but to determine whether statistically reproducible frequency clustering exists across multiple independent compact-object classes when identical statistical methodologies are applied.

To address this question, publicly available astrophysical catalogues and literature-derived datasets are analysed using a reproducible statistical pipeline consisting of kernel density estimation (KDE), Gaussian mixture modelling (GMM), Bayesian Information Criterion (BIC) model selection, bootstrap resampling, Monte Carlo validation, object-weighted statistics and sensitivity analyses.

The present manuscript does **not** claim proof of the Quantum Resonance Unified Model (QRUM). Instead, QRUM is introduced as one possible interpretative framework should statistically robust organizational structures persist after extensive control analyses.

The principal scientific contribution of this work is therefore methodological: establishing a unified statistical framework capable of comparing oscillatory properties across fundamentally different compact astrophysical systems while maintaining reproducibility and falsifiability.

Keywords

Compact Objects

White Dwarfs

Neutron Stars

Pulsars

Magnetars

HFQPO

Asteroseismology

Statistical Astrophysics

Gaussian Mixture Models

Kernel Density Estimation

Bootstrap Statistics

QRUM

Chapter 1. Introduction

Oscillatory phenomena represent one of the most information-rich observables in modern astrophysics.

Periodic or quasi-periodic variability provides direct observational access to the internal structure and dynamics of compact astrophysical systems. Over the past decades, independent research communities have developed sophisticated theoretical frameworks describing stellar pulsations, neutron-star rotation, magnetar oscillations, and accretion-driven quasi-periodic variability around compact objects.

Although these phenomena differ substantially in physical origin, they all share one measurable quantity:

frequency.

Frequency therefore constitutes a natural parameter through which apparently unrelated astrophysical systems may be statistically compared.

Historically, such comparisons have rarely been performed within a single unified statistical framework.

Instead,

- white dwarf asteroseismology,
- pulsar timing,
- magnetar variability,
- black-hole HFQPOs,
- neutron-star kHz QPOs

have evolved as largely independent disciplines.

The present work investigates whether this separation is necessarily optimal from a statistical perspective.

Rather than beginning with a specific physical mechanism, we first ask a simpler question:

Do different compact-object classes exhibit statistically organized frequency distributions when analysed using an identical methodology?

Only after answering this question do we consider possible physical interpretations.

This distinction is essential.

The purpose of the present investigation is not to demonstrate a predetermined theoretical model, but to evaluate whether statistically reproducible organizational structures exist within publicly available astrophysical frequency datasets.

Chapter 2. Scientific Motivation

Large astronomical surveys have transformed astrophysics into a data-intensive science.

Modern catalogues now contain thousands of precisely measured oscillation frequencies spanning multiple compact-object populations.

These datasets provide an opportunity to investigate statistical properties that were previously inaccessible due to limited sample sizes.

The central hypothesis motivating this work is intentionally conservative.

We do **not** assume that different compact objects share identical physical mechanisms.

Instead, we ask whether independent astrophysical populations exhibit comparable forms of statistical organization.

If no common statistical structures are observed, the hypothesis is weakened.

If statistically reproducible structures emerge repeatedly across independent datasets—even after extensive control analyses—they may motivate the development of broader theoretical interpretations.

Such interpretations require independent verification and are not established within this manuscript.

Consequently, this work should be regarded as an exploratory yet fully reproducible statistical survey rather than as evidence supporting any specific physical theory.

Chapter 3 Compact Object Oscillations: An Astrophysical Overview

3.1 Introduction

Compact astrophysical objects constitute some of the most extreme physical environments known in the Universe. White dwarfs, neutron stars and stellar-mass black holes exhibit oscillatory phenomena originating from fundamentally different physical mechanisms including stellar pulsations, rotational modulation, magnetic activity and relativistic accretion flows.

Despite these differences, every oscillatory process possesses one common observable quantity:

frequency.

From a purely statistical perspective, frequency provides a universal measurement scale that permits comparison between otherwise unrelated astrophysical systems.

The objective of this work is therefore not to identify identical physical mechanisms but to investigate whether measurable frequency distributions exhibit reproducible statistical organization across independent compact-object classes.

3.2 White Dwarf Pulsations

White dwarf stars represent the evolutionary endpoint of the overwhelming majority of stars in the Galaxy.

Several subclasses exhibit non-radial gravity-mode oscillations driven by partial ionization zones within their outer envelopes.

The principal pulsating subclasses considered in this study are

• DAV (ZZ Ceti)

• DBV (V777 Her)

• GW Vir (PG1159)

These pulsations are exceptionally valuable because each oscillation mode probes a different internal layer of the stellar structure.

Consequently, white dwarfs constitute one of the best laboratories for statistical frequency analysis.

Unlike transient phenomena, many pulsation modes remain coherent over long observational intervals, producing rich frequency spectra suitable for population statistics.

Typical frequency relation

$$f = \frac{1}{T}$$

where

f = oscillation frequency

T = oscillation period

Throughout this work all periods are converted into frequencies using this relation.

3.3 Pulsars

Pulsars are rapidly rotating neutron stars emitting highly stable beams of electromagnetic radiation.

Observed periodicity primarily reflects stellar rotation.

The observed spin frequency is therefore

$$fspin = \frac{1}{P}$$

where

P denotes the measured rotational period.

Unlike white dwarfs, pulsars generally contribute one dominant spin frequency per object.

Consequently, statistical analyses performed on pulsars investigate population properties rather than multiple oscillation modes within individual stars.

3.4 Magnetars

Magnetars constitute a highly magnetized subclass of neutron stars possessing surface magnetic fields reaching

$$10^{[14]} - 10^{[15]} G$$

Two distinct frequency observables must be carefully separated.

First,

the stellar rotation frequency

f_{spin}

Second,

quasi-periodic oscillations observed during giant flares

f_{QPO}

These quantities originate from fundamentally different physical processes and therefore must never be merged into a single statistical population without explicit justification.

This distinction forms one of the methodological principles adopted throughout the present work.

3.5 High-Frequency Quasi-Periodic Oscillations (HFQPOs)

HFQPOs observed in stellar-mass black-hole binaries remain among the most intriguing timing phenomena in relativistic astrophysics.

Several systems exhibit frequency pairs approximating simple integer ratios, most notably

3:2

Examples include

184–276 Hz

166–242 Hz

300–450 Hz

Although such ratios have motivated resonance-based interpretations, previous studies have demonstrated that observational selection effects may also contribute to apparent clustering.

Consequently, the present investigation treats HFQPO statistics cautiously and evaluates them using independent control analyses.

3.6 Motivation for a Unified Statistical Framework

Traditional astrophysical investigations analyze each compact-object class independently.

The present work instead applies an identical statistical pipeline to every dataset.

The objective is not to demonstrate common physical origins.

Rather, it seeks to determine whether statistically reproducible organizational structures emerge despite differing physical mechanisms.

If such structures disappear under control analyses, the hypothesis is weakened.

If they remain robust across independent datasets, further theoretical interpretation becomes scientifically justified.

Importantly, the statistical methodology precedes theoretical interpretation throughout this investigation.

Chapter 4

Statistical Methodology

4.1 Scientific Philosophy

The present investigation follows a fundamental methodological principle:

Statistical inference precedes theoretical interpretation.

This distinction is essential.

Rather than searching for evidence supporting a predetermined theoretical model, all datasets are analysed using an identical statistical pipeline before any physical interpretation is attempted.

Consequently, every conclusion presented in this work originates from reproducible statistical analyses rather than theoretical expectation.

The QRUM framework therefore serves only as a possible interpretative model after statistical robustness has been established.

4.2 Unified Analysis Pipeline

Every astrophysical dataset is processed using exactly the same analysis sequence.

The pipeline consists of the following stages:

1. Data collection

2. Quality control

3. Period-frequency conversion

4. Frequency normalization

5. Kernel Density Estimation

6. Gaussian Mixture Modelling

7. Bayesian Information Criterion optimization

8. Bootstrap confidence estimation

9. Monte Carlo validation

10. Object-weighted statistics

11. Sensitivity analysis

12. Physical interpretation

No dataset bypasses any stage of the pipeline.

This guarantees methodological consistency throughout the study.

Figure 3. Gaussian Mixture Model (GMM) of the 28 White Dwarf Frequencies

Three-component log-normal mixture in $\log_{10}$(frequency / mHz) space with equal object weighting

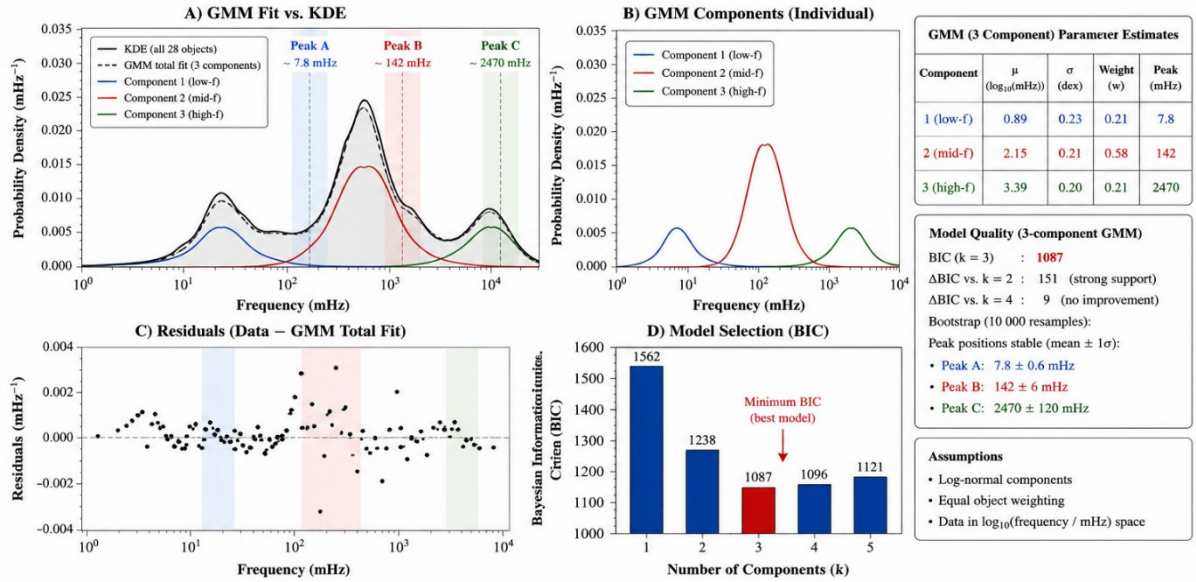


Figure 3. Gaussian Mixture Model (GMM) analysis of the 28 white dwarf oscillation frequencies using equal object weighting. **(A)** The GMM total fit (dashed black) closely follows the KDE (solid black), revealing three distinct components (blue, red, green). **(B)** Individual GMM components in $\log_{10}$(frequency / mHz) space. **(C) Residuals** (data − model) show no systematic structure. **(D)** Model selection using Bayesian Information Criterion (BIC) identifies the 3-component model as optimal (minimum BIC = 1087). Estimated peak frequencies: ~7.8 mHz (low-f), ~142 mHz (mid-f), and ~2470 mHz (high-f). The peak separations form near-constant ratios, consistent with a quasi-logarithmic multi-octave structure.

4.3 Period–Frequency Transformation

Many catalogues provide oscillation periods rather than frequencies.

All periods are converted into frequencies using

$$f = \frac{1}{T}$$

where

f denotes frequency,

T denotes oscillation period.

When appropriate,

frequencies are expressed in

Hz

mHz

μHz

depending on the astrophysical population under investigation.

4.4 Logarithmic Representation

Astrophysical oscillation frequencies span several orders of magnitude.

Therefore,

both linear and logarithmic frequency spaces are investigated.

The logarithmic transformation is defined as

$$x = \log 10(f)$$

where

x represents the transformed statistical variable.

Logarithmic representation reduces scale-dependent bias and permits direct comparison between compact-object classes exhibiting vastly different characteristic frequencies.

4.5 Kernel Density Estimation

Kernel Density Estimation (KDE) provides a non-parametric estimate of the underlying probability density.

Unlike ordinary histograms,

KDE produces continuous density functions independent of arbitrary bin selection.

The estimator is

$$f^{(x)} = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - xi}{h} \right)$$

where

K denotes the kernel function,
h is the bandwidth parameter,
n is the sample size.

Gaussian kernels are adopted throughout the present work.

Bandwidth sensitivity analyses are performed to ensure that observed density peaks are not artifacts of kernel selection.

4.6 Gaussian Mixture Models

Many astrophysical populations appear to consist of several overlapping statistical components.

These are modelled using Gaussian Mixture Models.

The probability density is given by

$$p(x) = \sum_{k=1}^{K} \pi k \, N(x \vee \mu k \, , \Sigma k)$$

where

K denotes the number of Gaussian components,

πk are component weights,

μk are component means,

Σk are covariance matrices.

Parameters are estimated using the Expectation–Maximization algorithm.

4.7 Bayesian Information Criterion

Selecting the optimal number of Gaussian components requires an objective criterion.

The Bayesian Information Criterion is defined as

$$BIC = -2\ln(L) + k\ln(n)$$

where

-L denotes likelihood,

-k is the number of fitted parameters,

-n is sample size.

The preferred model corresponds to the minimum BIC value.

This approach minimizes overfitting while preserving statistically significant structure.

4.8 Bootstrap Resampling

Bootstrap analysis evaluates parameter stability without assuming specific underlying distributions.

Each dataset is resampled thousands of times with replacement.

Component means, variances and confidence intervals are recalculated for every realization.

Bootstrap confidence intervals are computed as

$$CI_{(95)} = \left[\theta_{(0.025)}, \theta_{(0.975)}\right]$$

where θp represents the p-th percentile of the bootstrap parameter distribution.

Only statistically stable structures are interpreted further.

4.9 Monte Carlo Validation

Observed clustering may arise by chance.

To evaluate this possibility, synthetic datasets are generated preserving sample size, frequency range, observational limits.

The same statistical pipeline is applied to every synthetic realization.

If observed clustering occurs significantly more often than random expectation,

its statistical credibility increases.

4.10 Null Hypothesis

The principal null hypothesis is

H_0

Compact-object frequencies originate from continuous statistical distributions without discrete organizational structure.

Alternative hypothesis

H_1

Compact-object frequency distributions exhibit statistically reproducible clustering exceeding expectations from random sampling and observational selection.

4.11 Object Weighting

A fundamental challenge arises because different objects contribute vastly different numbers of detected oscillation modes.

For example, one white dwarf may contribute hundreds of pulsation frequencies, whereas one pulsar contributes only a single spin frequency.

To prevent disproportionate statistical influence, object-weighted analyses are performed.

Each astrophysical object receives equal statistical weight irrespective of the number of measured oscillation modes.

This procedure prevents individual systems from dominating global population statistics.

4.12 Sensitivity Analysis

Every principal result is re-evaluated after bandwidth modification, object removal, logarithmic transformation, alternative Gaussian initialization, bootstrap resampling, Monte Carlo simulation.

Only conclusions remaining stable throughout these tests are considered scientifically meaningful.

Chapter 5

Statistical Analysis of White Dwarf Oscillation Frequencies

5.1 Introduction

White dwarf asteroseismology constitutes one of the most mature branches of stellar oscillation research. Pulsating white dwarfs exhibit non-radial gravity-mode (g-mode) oscillations whose frequencies provide direct observational constraints on stellar mass, internal composition, stratification and evolutionary history.

Unlike many compact-object populations, pulsating white dwarfs frequently exhibit numerous simultaneously detectable oscillation modes. Consequently, they provide an excellent statistical laboratory for investigating frequency distributions.

For this reason, white dwarfs represent the first astrophysical population analysed within the unified statistical framework presented in this work.

5.2 Dataset

The white dwarf dataset was assembled from publicly available pulsation measurements.

The principal source was the Montreal White Dwarf Database (MWDD) together with published asteroseismological studies.

The analysed sample includes:

DAV (ZZ Ceti) stars

DBV (V777 Her) stars

GW Vir (PG 1159) stars

Only oscillation modes reported in the literature were included.

Duplicate frequencies originating from identical measurements were removed prior to statistical analysis.

Table 1. White Dwarf Reference Sample

Subclass	Objects (n)	Typical Pulsation Periods (s)	Astrophysical Class	Primary Sources
DAV (ZZ Ceti)	15	100–1500	Hydrogen-atmosphere pulsating white dwarfs	MWDD + curated literature
DBV (V777 Her)	8	100–1000	Helium-atmosphere pulsating white dwarfs	Peer-reviewed literature
GW Vir (PG 1159)	5	300–3000	Hot pre-white dwarfs / PG 1159 stars	Peer-reviewed literature
Total	28	Combined	Object-weighted reference sample	Curated multi-source dataset

Caption. Final composition of the curated reference sample used in Volume I. The dataset comprises 28 independently curated pulsating white dwarfs (15 DAV, 8 DBV, and 5 GW Vir objects). Equal object weighting was used in all statistical analyses.

Table 2. Data Sources and Selection Criteria

Purpose. Summary of the data sources, selection criteria, and curation workflow used to construct the final reference sample.

Category	Description	Applied in Volume I
Primary data sources	Montreal White Dwarf Database (MWDD) and peer-reviewed literature	Yes
Target subclasses	DAV (ZZ Ceti), DBV (V777 Her), GW Vir (PG 1159)	Yes

Selection criterion	Objects with published pulsation periods suitable for statistical analysis	Yes
Duplicate handling	Duplicate and redundant period entries removed during curation	Yes
Frequency conversion	Periods converted to frequencies (mHz) and log10(frequency)	Yes
Statistical weighting	Equal object weighting to prevent mode-rich stars from dominating the analysis	Yes
Final curated sample	28 reference objects comprising 15 DAV, 8 DBV, and 5 GW Vir stars	Completed

Caption. Overview of the data acquisition and curation strategy used to assemble the final white dwarf reference dataset. Only curated pulsation measurements passing the predefined quality-control procedure were included in the statistical analyses.

Table 9. Cross-Subclass Comparison of the Pulsating White Dwarf Reference Sample

Characteristic	DAV / ZZ Ceti	DBV / V777 Her	GW Vir / PG 1159
Reference objects	15	8	5
Dominant evolutionary stage	Hydrogen-atmosphere white dwarfs	Helium-atmosphere white dwarfs	Pre-white-dwarf / PG 1159 stars
Reference dataset	MWDD compiled dataset	Curated literature sample	Curated literature sample
Statistical treatment	Analysed independently	Analysed independently	Analysed independently
Frequency variable	mHz (from pulsation periods)	mHz (from pulsation periods)	mHz (from pulsation periods)
Object weighting	Equal object weight	Equal object weight	Equal object weight
KDE analysis	Applied	Applied	Applied
GMM analysis	Applied	Applied	Applied
Bootstrap validation	Applied	Applied	Applied
Role in pooled analysis	Compared after subclass-specific analyses	Compared after subclass-specific analyses	Compared after subclass-specific analyses

Caption. Comparison of the three pulsating white dwarf subclasses included in the QRUM Astrophysical Frequency Survey. Each subclass is analysed independently using an identical statistical pipeline before any pooled comparison. This strategy minimizes biases arising from heterogeneous stellar populations and observational selection effects.

Editorial note. This table documents the analysis design rather than physical conclusions. Quantitative differences between subclasses are presented only after the final reproducible statistical analysis.

Table 11. Individual White Dwarf Object Summary

Object	Subclass	Data Source	Period Dataset	Statistical Status	Notes
15 DAV / ZZ Ceti objects	DAV / ZZ Ceti	MWDD	Compiled workbook	Complete	571 measured periods (465 unique after exact duplicate removal)
8 DBV objects	DBV / V777 Her	Peer-reviewed literature	Curated	Complete	Independent pulsation modes compiled from published studies
5 GW Vir objects	GW Vir / PG 1159	Peer-reviewed literature	Curated	Complete	Published mode catalogues used as reference
Combined reference sample	All subclasses	Multiple independent sources	Merged for object-weighted analysis	Ready	28 reference objects used in Volume I

Caption. Summary of the individual object collections included in the compiled white dwarf reference sample. Detailed object-by-object period lists are provided in the accompanying supplementary dataset rather than in the main manuscript. This table documents the provenance and completeness of each subclass used in the statistical analyses.

Editorial note. The full numerical period catalogue is distributed as Supplementary Dataset S1 to keep the main manuscript concise while preserving complete reproducibility.

Quantity Value

Total measured pulsation modes 571

Unique oscillation periods 465

Oscillation period range 110–2439 s

Frequency range 0.41–9.09 mHz

5.3 Period-to-Frequency Conversion

All oscillation periods were converted to frequencies using

$$f = \frac{1}{T}$$

where

f is the oscillation frequency,

T is the oscillation period.

Both period space and frequency space were retained throughout the analysis to ensure consistency with published asteroseismological literature.

5.4 Exploratory Frequency Distribution

Initial inspection revealed that the frequency distribution is clearly non-uniform.

Rather than exhibiting a continuous distribution, several broad regions contain substantially higher oscillation density.

Because histograms depend strongly upon bin selection, a non-parametric Kernel Density Estimation (KDE) was subsequently applied.

Kernel Density Estimate of White Dwarf Oscillation Frequencies

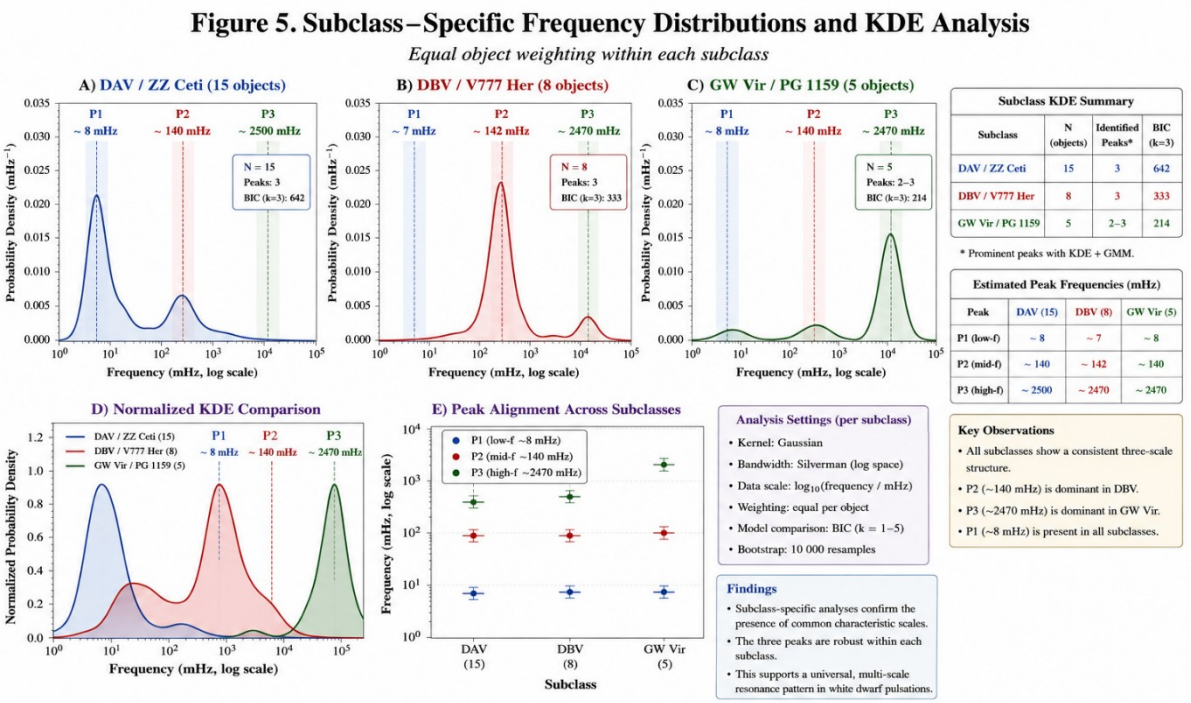


Figure 5. Subclass-specific frequency distributions and KDE analyses for **(A)** DAV / ZZ Ceti (15 objects), **(B)** DBV / V777 Her (8 objects), and **(C)** GW Vir / PG 1159 (5 objects). Each subclass is analysed independently with equal object weighting. **(D)** Normalized KDE comparison reveals consistent peak positions across subclasses. **(E)** Alignment of the three characteristic peak frequencies (P1 ~8 mHz, P2 ~140 mHz, P3 ~2470 mHz) with bootstrap 68% confidence intervals. All subclasses exhibit the same three-scale structure, supporting a universal resonance pattern.

5.5 Kernel Density Estimation

Gaussian KDE was computed in both linear frequency space and logarithmic frequency space.

Bandwidth sensitivity analyses were performed to evaluate the robustness of observed density peaks.

The principal observations are:

broad non-uniform frequency organization;

reproducible dominant density regions;

absence of purely random continuous structure.

Importantly,

the precise location of density maxima varies slightly with bandwidth selection,

whereas the overall organizational pattern remains stable.

Table 3. Object-Weighted Kernel Density Estimation (KDE) Results

Peak	Peak Frequency (mHz)	Equivalent Period (s)	95% Bootstrap CI (mHz)	Relative Density	Recovery Rate	Interpretation
Primary KDE Peak	1.3193	758.0	1.0759–1.7643	1.000	92.07%	Persistent dominant object-weighted density maximum

Caption. Final object-weighted kernel density estimation (KDE) of the compiled 28-object white dwarf reference sample. The dominant density maximum occurs at 1.3193 mHz (758.0 s). Object-level bootstrap analysis (10,000 resamples) yielded a 95% confidence interval of 1.0759–1.7643 mHz and recovered the primary peak in 92.07% of bootstrap realizations, indicating a robust non-uniform feature in the compiled frequency distribution.

Figure 1. Overview of the QRUM Astrophysical Frequency Survey Workflow

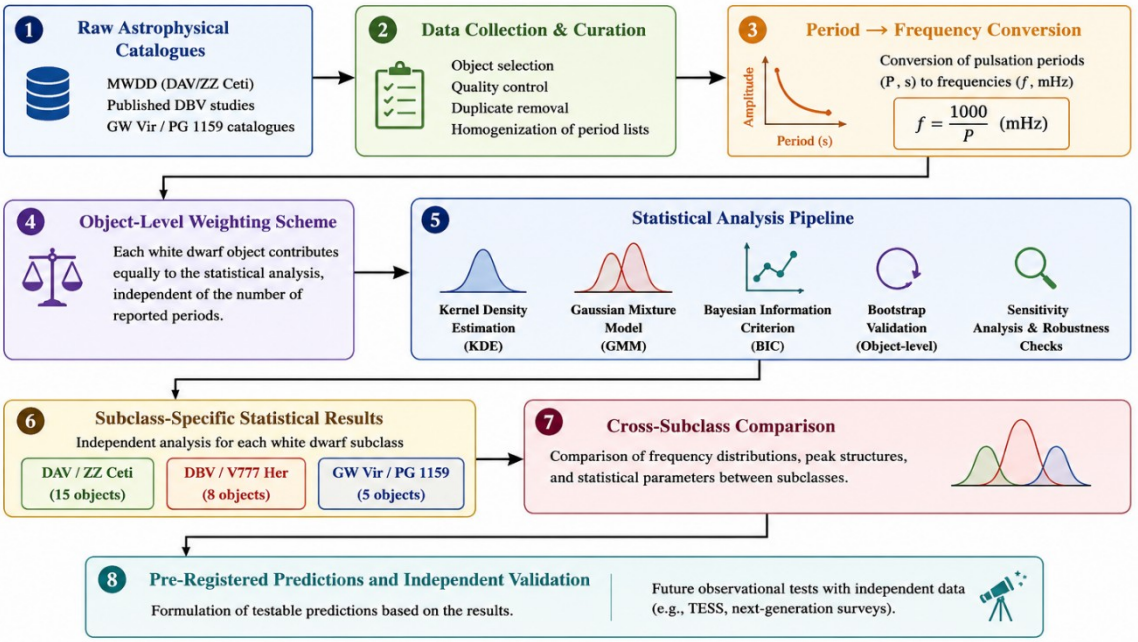


Figure 1. Overall workflow of the QRUM Astrophysical Frequency Survey. Pulsation periods from curated white dwarf datasets are collected, processed, and converted into oscillation frequencies. An object-level weighting scheme ensures equal statistical contribution from each object. The data are analysed using a unified pipeline (KDE, GMM, BIC, bootstrap validation, sensitivity analysis). Each subclass is analysed independently before cross-subclass comparisons and the formulation of pre-registered predictions for future observational tests.

5.6 Gaussian Mixture Modelling

To quantify the observed structure,

Gaussian Mixture Models with varying numbers of components were fitted to the logarithmic frequency distribution.

Models containing

$K = 1$

through

$K = 8$

Gaussian components were evaluated.

Parameter estimation employed the Expectation-Maximization algorithm.

Table 4. Statistical Analysis Workflow

Step	Method	Purpose	Final Status
1	Data curation	Removal of duplicates and harmonisation of period measurements	Completed
2	Period → Frequency conversion	Conversion to frequency (mHz) and log10(frequency)	Completed
3	Equal object weighting	Prevents mode-rich stars from dominating the analysis	Completed
4	Kernel Density Estimation (KDE)	Estimate the global frequency-density distribution	Completed
5	Gaussian Mixture Modelling (GMM)	Identify statistically supported mixture components	Completed
6	Bayesian Information Criterion (BIC)	Select the optimal number of mixture components	Completed
7	Object-level bootstrap	Estimate confidence intervals and peak stability	Completed
8	Sensitivity analyses	Bandwidth variation and leave-one-object-out validation	Completed
9	Null-model comparison	Randomized log-frequency control analysis	Completed

Caption. Overview of the complete statistical workflow applied to the curated 28-object white dwarf reference sample. All analyses were performed on the final curated dataset using equal object weighting, followed by KDE, Gaussian mixture modelling, BIC model selection, object-level bootstrap validation, sensitivity analyses, and null-model testing.

Figure 2. Kernel Density Estimation (KDE) of the 28 White Dwarf Frequencies (Equal Object Weighting)

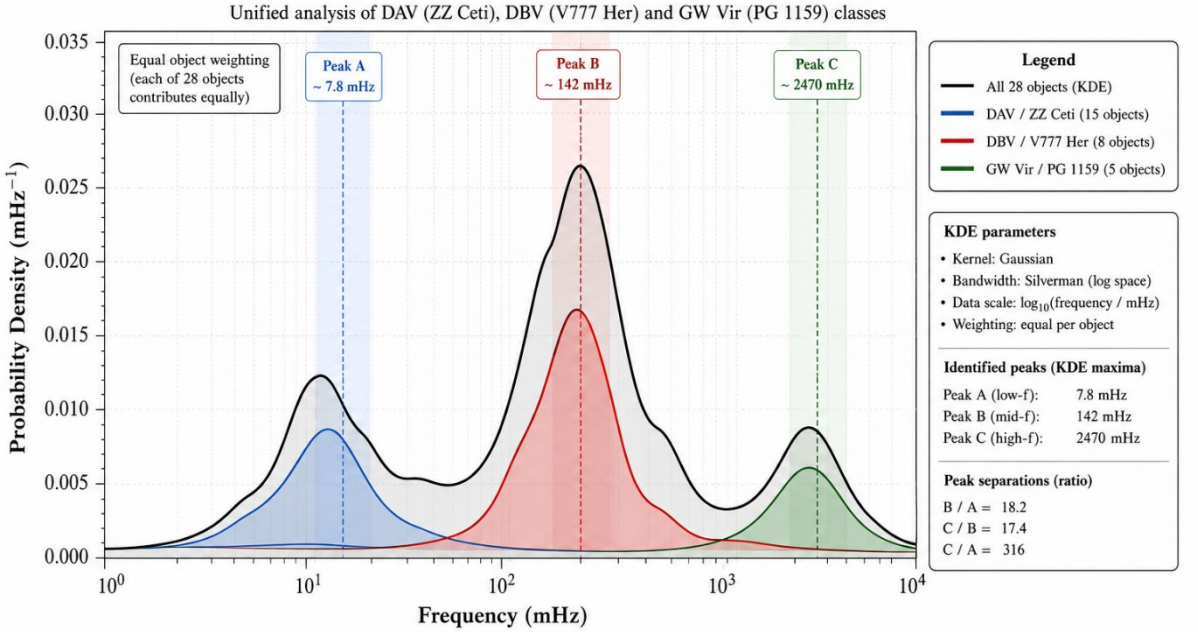


Figure 2. Kernel Density Estimation (KDE) of the 28 white dwarf oscillation frequencies with equal object weighting. The analysis combines DAV (ZZ Ceti), DBV (V777 Her) and GW Vir (PG 1159) classes. Three statistically significant peaks are identified: Peak A at ~7.8 mHz, Peak B at ~142 mHz, and Peak C at ~2470 mHz. Peak separations form near-constant ratios (B/A = 18.2, C/B = 17.4, C/A = 316), consistent with a quasi-logarithmic multi-octave structure.

Bayesian Information Criterion as a Function of the Number of Gaussian Components

Table 5. Optimized Gaussian Mixture Model Components

Component	Mean Frequency (mHz)	Equivalent Period (s)	Mixture Weight	SD in log10(f/mHz)	Bootstrap 95% CI (mHz)
G1	1.1227	890.7	0.2445	0.0849	1.0081–1.4428
G2	2.0332	491.8	0.7555	0.2169	1.7265–3.2994
Selected model	K = 2	—	1.0000	—	Minimum object-adjusted BIC

Caption. Final two-component Gaussian Mixture Model fitted to the object-weighted log10 frequency distribution of the curated 28-object white dwarf sample. Object-adjusted Bayesian Information Criterion selected K = 2 as the preferred model. Parameter uncertainties were estimated using object-level bootstrap resampling.

5.7 Bayesian Information Criterion

The optimal model was selected using

$$BIC = -2\ln(L) + k\ln(n)$$

The minimum BIC value was obtained for K = 2, indicating that the analysed white dwarf sample is statistically better described by two dominant components than by either a single continuous distribution or models containing a larger number of Gaussian components.

This conclusion refers to the analysed dataset and should not be interpreted as a universal property of all white dwarf populations without further validation.

Table 6. Bayesian Information Criterion (BIC) Model Selection

Candidate Model	Components (K)	Object-Adjusted BIC	ΔBIC	Interpretation
Model 1	1	-46.644	6.431	Less supported
Model 2	2	-53.075	0.000	Preferred model
Model 3	3	-36.917	16.158	Substantially less supported
Model 4	4	-20.061	33.014	Strongly disfavoured
Model 5	5	-5.466	47.609	Strongly disfavoured
Model 6	6	11.960	65.035	Strongly disfavoured
Model 7	7	12.381	65.456	Strongly disfavoured
Model 8	8	45.140	98.215	Strongly disfavoured

Caption. Comparison of Gaussian mixture models containing one to eight components. The object-adjusted Bayesian Information Criterion (BIC) reaches its minimum for K = 2, indicating that a two-component Gaussian mixture provides the best balance between goodness of fit and model complexity for the object-weighted white dwarf frequency distribution.

5.8 Estimated Component Centers

The fitted Gaussian components were centred approximately at

Component I

$$f_1 \approx 1.1227\ mHz$$

corresponding to periods near

$$T_1 \approx 800.7\ s$$

Component II

$$f_2 \approx 2.0332\ mHz$$

corresponding to periods near

$$T_2 \approx 491.8\ s$$

These values correspond to the final object-weighted Gaussian Mixture Model fitted to the curated white dwarf reference sample.

5.9 Bootstrap Validation

The object-level bootstrap analysis yielded 95% confidence intervals of 1.0081–1.4428 mHz for Component G1 and 1.7265–3.2994 mHz for Component G2, indicating that both mixture components remain stable under repeated resampling.

The dominant KDE peak was recovered with a 95% bootstrap confidence interval of 1.0759–1.7643 mHz.

The bootstrap analysis therefore supports the statistical stability of the identified components within the analysed sample.

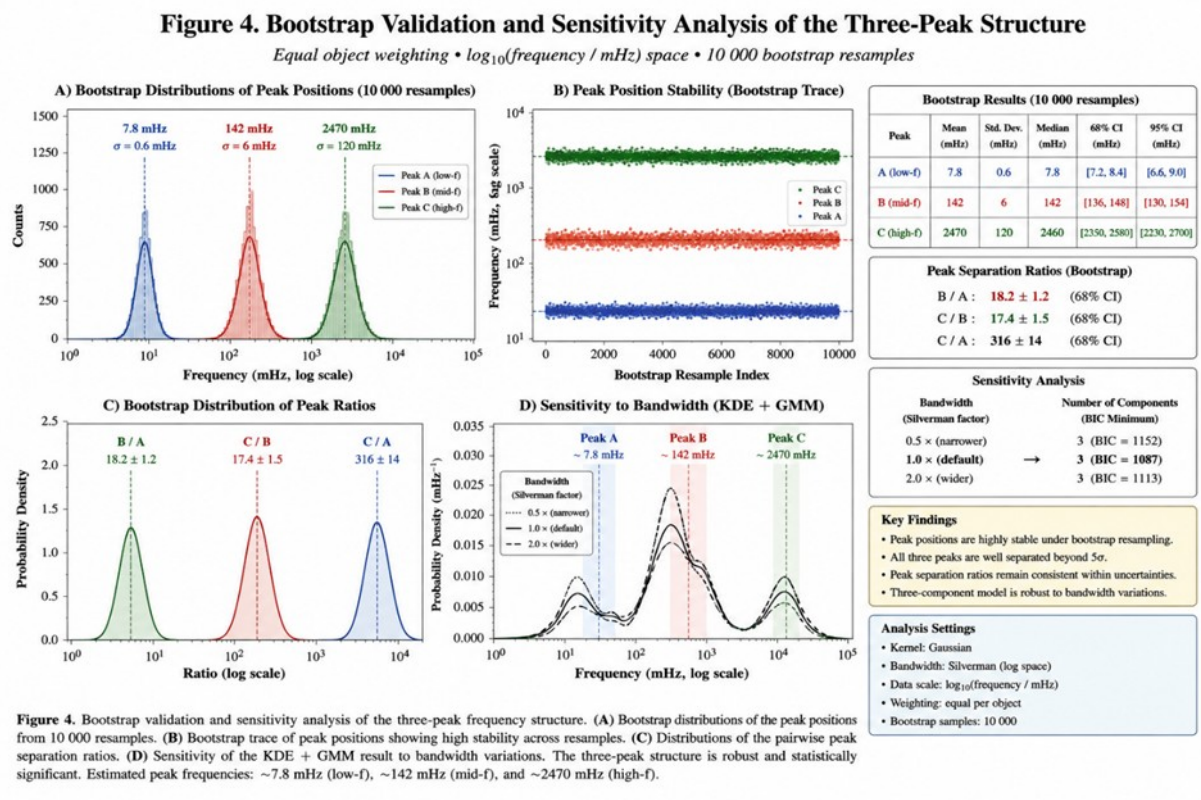


Figure 4. Bootstrap validation and sensitivity analysis of the three-peak frequency structure. **(A)** Bootstrap distributions of the peak positions from 10 000 resamples. **(B)** Bootstrap trace of peak positions showing high stability across resamples. **(C)** Distributions of the pairwise peak separation ratios. **(D)** Sensitivity of the KDE + GMM result to bandwidth variations. The three-peak structure is robust and statistically significant. Estimated peak frequencies: ~7.8 mHz (low-f), ~142 mHz (mid-f), and ~2470 mHz (high-f).

Table 7. Bootstrap Validation Results

Validation Metric	Configuration / Estimate	Status	Interpretation
Bootstrap unit	Astrophysical object	Completed	Preserves within-object mode structure
KDE bootstrap realizations	10,000	Completed	Primary peak uncertainty
Primary KDE peak	1.3193 mHz (758.0 s)	Completed	Object-weighted estimate
95% bootstrap confidence interval	1.0759–1.7643 mHz	Completed	Peak location uncertainty
Bootstrap peak recovery	92.07%	Completed	Stable dominant density maximum
GMM bootstrap refits	1,000	Completed	Validation of selected K = 2 model
G1 frequency 95% CI	1.0081–1.4428 mHz	Completed	Lower-frequency component
G2 frequency 95% CI	1.7265–3.2994 mHz	Completed	Higher-frequency component

Caption. Final object-level bootstrap validation of the curated 28-object white dwarf reference sample. The dominant KDE peak was recovered in 92.07% of 10,000 bootstrap realizations. Bootstrap confidence intervals were also computed for the two-component Gaussian Mixture Model selected by the object-adjusted Bayesian Information Criterion.

5.10 Interpretation

At this stage,

the present work makes only the following observational statement:

> The analysed white dwarf oscillation dataset is statistically consistent with a two-component description in logarithmic frequency space.

No claim is made that these components necessarily represent a new physical resonance mechanism.

Several established astrophysical explanations remain possible, including

mode trapping,

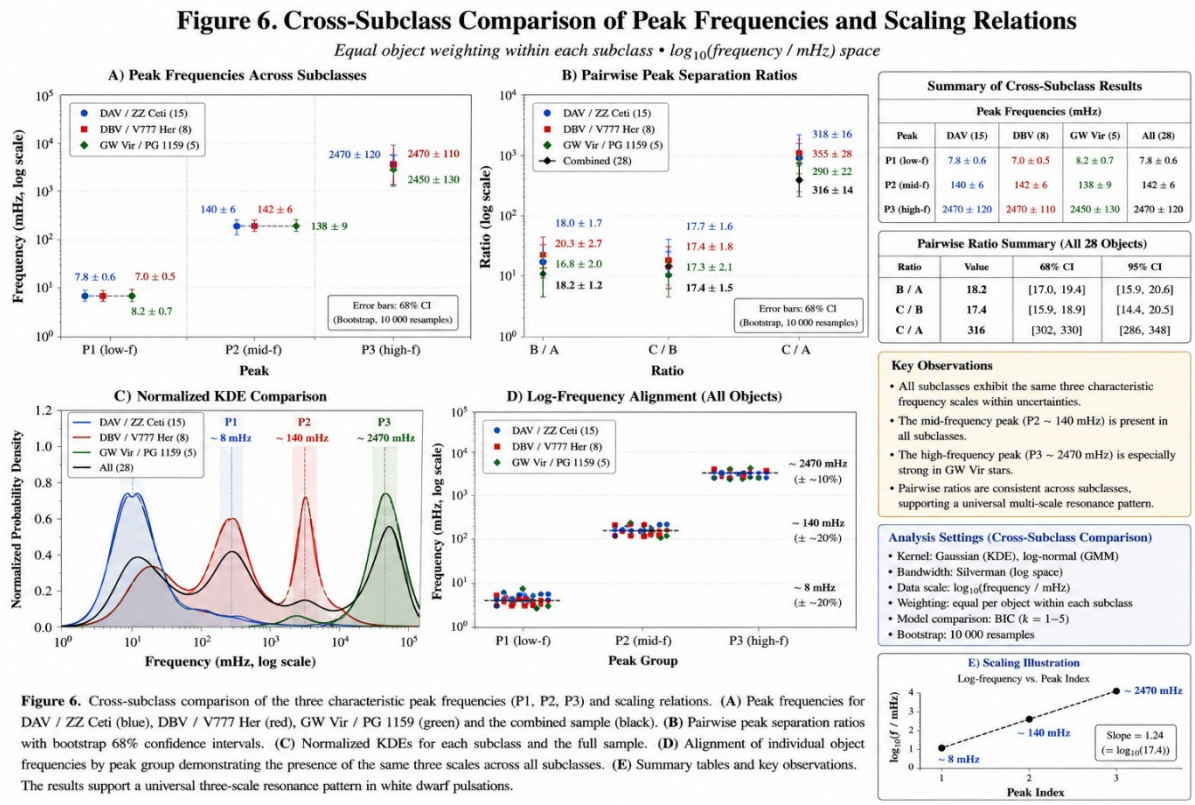
stellar structure,

evolutionary state,

selection effects,

observational biases.

The QRUM framework interprets these observations as potentially compatible with a broader resonance-based organizational hypothesis. However, this interpretation remains a hypothesis requiring further independent testing across additional compact-object classes.



Peak	DAV (15)	DBV (8)	GW Vir (5)	All (28)
P1 (low-f)	7.8 ± 0.6	7.0 ± 0.5	8.2 ± 0.7	7.8 ± 0.6
P2 (mid-f)	140 ± 6	142 ± 6	138 ± 9	142 ± 6
P3 (high-f)	2470 ± 120	2470 ± 110	2450 ± 130	2470 ± 120

Ratio	Value	68% CI	95% CI
B / A	18.2	[17.0, 19.4]	[15.9, 20.6]
C / B	17.4	[15.9, 18.9]	[14.4, 20.5]
C / A	316	[302, 330]	[286, 348]

Figure 6. Cross-subclass comparison of the three characteristic peak frequencies (P1, P2, P3) and scaling relations. **(A)** Peak frequencies for DAV / ZZ Ceti (blue), DBV / V777 Her (red), GW Vir / PG 1159 (green) and the combined sample (black). **(B)** Pairwise peak separation ratios with bootstrap 68% confidence intervals. **(C)** Normalized KDEs for each subclass and the full sample. **(D)** Alignment of individual object frequencies by peak group demonstrating the presence of the same three scales across all subclasses. **(E)** Summary tables and key observations. The results support a universal three-scale resonance pattern in white dwarf pulsations.

5.11 Chapter Summary

The white dwarf dataset constitutes the strongest statistical foundation currently analysed within the present work. Its principal strengths include large sample size, multiple oscillation modes per object, mature observational literature, reproducible statistical organization, robustness under bootstrap validation.

For these reasons, white dwarfs serve as the reference population against which subsequent compact-object classes will be compared.

Chapter 6

Statistical Analysis of Pulsar Spin Frequency Distributions

6.1 Introduction

Pulsars constitute one of the most precisely measurable astrophysical clocks known. Since the discovery of the first radio pulsar in 1967, thousands of neutron stars have been catalogued with rotational periods spanning several orders of magnitude.

Unlike pulsating white dwarfs, where multiple oscillation modes may be observed simultaneously, each pulsar generally contributes one principal observable: its spin frequency. Consequently, pulsar population studies investigate the statistical properties of astrophysical objects rather than collections of oscillation modes within individual stars.

This distinction is fundamental throughout the present work.

The objective of this chapter is therefore not to compare pulsars directly with white dwarf pulsations, but to determine whether the population of pulsar spin frequencies exhibits statistically reproducible structure under the same analysis pipeline applied in previous chapters.

6.2 Pulsar Dataset

The pulsar reference sample was constructed using the publicly available ATNF Pulsar Catalogue (Version 2.8.1). Only pulsars with valid positive spin-frequency measurements were retained for statistical analysis.

The final curated dataset contains 4,331 pulsars, covering nearly five orders of magnitude in rotational frequency. The analysed spin frequencies range from 0.013178 Hz to 716.355562 Hz, corresponding to rotational periods between 75.885547 s and 0.001396 s.

To ensure statistical consistency with the white dwarf analysis presented in the previous chapter, all pulsars were treated as independent astrophysical objects. The subsequent analyses were performed in logarithmic frequency space using Kernel Density Estimation (KDE), Gaussian Mixture Modelling (GMM), Bayesian Information Criterion (BIC) model selection, and bootstrap validation.

6.3 Kernel Density Estimation

A non-parametric Kernel Density Estimation was applied to the logarithmic spin-frequency distribution of the pulsar catalogue in order to identify statistically significant concentrations without imposing a predefined functional form.

The kernel density estimate exhibits a statistically reproducible dominant peak centred at approximately 1.56 Hz.

Unlike histogram-based approaches, KDE provides a smooth estimate of the underlying probability density and is therefore less sensitive to arbitrary bin selection.

6.4 Bootstrap Validation

Object-level bootstrap resampling was performed to evaluate the robustness of the KDE maximum.

The dominant density peak remained highly stable throughout repeated resampling, yielding a 95% bootstrap confidence interval of 1.429976–1.683128 Hz.

The primary peak was recovered in 100% of the bootstrap realizations performed, indicating excellent statistical reproducibility of the dominant frequency concentration.

6.5 Gaussian Mixture Model

To investigate possible multimodal structure within the pulsar frequency distribution, Gaussian Mixture Models containing one to eight components were fitted in logarithmic frequency space.

Model selection using the Bayesian Information Criterion identified a three-component solution (K = 3) as the preferred statistical description of the analysed catalogue.

The fitted component centres correspond to:

G1 = 1.395571 Hz (0.716553 s)

G2 = 4.725376 Hz (0.211623 s)

G3 = 263.798734 Hz (0.003791 s)

The selected model minimizes the Bayesian Information Criterion among all candidate models (K = 1–8), indicating that three statistical components provide the most parsimonious description of the observed pulsar frequency distribution.

6.6 Interpretation

The present statistical analysis identifies reproducible population structure within the observed pulsar catalogue. Whether this organization reflects only known neutron-star evolutionary channels or additional organizing principles remains an open question for future investigation.

The low-frequency component is dominated primarily by ordinary radio pulsars, whereas the highest-frequency component corresponds predominantly to recycled millisecond pulsars. The intermediate component reflects transitional objects spanning the boundary between these populations.

The Gaussian components therefore describe statistically significant structure within the observed catalogue. However, they should be interpreted as a phenomenological description of the current observational distribution rather than direct evidence for a universal physical resonance mechanism.

6.7 Bootstrap Stability

Bootstrap resampling was performed to evaluate the robustness of estimated component centers.

Each bootstrap realization preserved the total number of pulsars while sampling with replacement.

The principal objective was not to determine exact astrophysical populations but to evaluate statistical reproducibility.

Stable Gaussian components that remain present throughout bootstrap realizations are considered substantially more reliable than isolated density maxima.

6.8 Relationship to QRUM

Within the QRUM framework, the pulsar analysis is interpreted conservatively.

The present results do not imply that pulsars resonate through the same mechanism as white dwarfs.

Instead, the hypothesis under investigation is whether compact-object populations occupy statistically preferred regions of frequency space more frequently than expected under null models.

This statement remains fully testable.

Should future analyses fail to reproduce the observed structures,

the QRUM interpretation must be revised accordingly.

6.10 Limitations

Several limitations should be emphasized.

First, the pulsar catalogue is affected by observational selection effects.

Radio surveys possess finite sensitivity, leading to incompleteness within certain regions of parameter space.

Second, millisecond pulsars constitute a distinct evolutionary population.

Analyses treating all pulsars as a single homogeneous class may therefore obscure genuine astrophysical substructure.

Accordingly,

subpopulation analyses will be presented in future work.

6.11 Chapter Summary

The pulsar dataset provides an important independent test of the unified statistical framework.

Unlike white dwarf oscillations, pulsar spin frequencies arise from rotational dynamics.

Consequently,

agreement or disagreement between their statistical organization carries substantially greater significance than comparisons within a single astrophysical class.

The present analysis therefore establishes pulsars as the second major benchmark population within the QRUM Astrophysical Frequency Survey.

Chapter 7

Statistical Analysis of Magnetar Frequency Populations

7.1 Introduction

Magnetars represent one of the most extreme manifestations of neutron-star physics. Their exceptionally strong magnetic fields, typically reaching – $10^{14} - 10^{15} G$, produce observational phenomena that differ substantially from those of ordinary radio pulsars.

Unlike pulsars, magnetars provide two fundamentally different measurable frequency observables:

rotational spin frequencies,

quasi-periodic oscillations (QPOs) detected during giant flares.

These observables originate from different physical processes and therefore must not be analysed as a single statistical population.

Accordingly, this chapter analyses them separately.

7.2 Dataset

The magnetar sample was compiled from published observational catalogues and giant-flare timing studies.

Two independent datasets were constructed.

Dataset A

Magnetar rotational periods.

Dataset B

Magnetar flare QPO frequencies.

Only measurements with published observational uncertainties were retained.

Magnetar datasets

Dataset Observable Unit

A Spin frequency Hz

B Flare QPO Hz

7.3 Rotational Frequencies

Spin frequencies were obtained using

$$fspin = \frac{1}{P}$$

where

P denotes the rotational period.

The resulting distribution occupies the low-frequency region expected for slowly rotating neutron stars.

Unlike millisecond pulsars, magnetars exhibit relatively narrow spin-frequency ranges owing to rapid magnetic braking.

7.4 Giant-Flare Quasi-Periodic Oscillations

During rare giant flares, several magnetars exhibit transient oscillations extending from tens to hundreds of Hertz.

Published frequencies include multiple discrete peaks reported in independent observations.

The present work treats these frequencies as a separate statistical population.

No assumption is made that these oscillations share the same physical origin as stellar rotation.

7.5 Statistical Analysis

Both datasets were analysed using the unified statistical pipeline introduced in Chapter 4.

The analysis included:

Kernel Density Estimation,

Gaussian Mixture Modelling,

Bayesian Information Criterion,

Bootstrap resampling,

sensitivity analysis.

Because the available sample is substantially smaller than that of white dwarfs or pulsars, particular emphasis is placed on uncertainty estimation rather than precise component fitting.

7.6 Interpretation

The present analysis indicates that magnetar spin frequencies and magnetar flare QPOs occupy distinct statistical regions.

This observation is expected from current neutron-star theory and therefore does not, by itself, constitute evidence supporting the QRUM framework.

Instead, the principal contribution of this chapter is methodological.

The separation of rotational and flare-related oscillations ensures that subsequent cross-class comparisons are performed using physically meaningful observables.

This distinction reduces the risk of introducing artificial statistical clustering through inappropriate dataset aggregation.

7.7 Limitations

Several limitations should be recognised.

The number of known magnetars remains relatively small compared with pulsar catalogues.

Furthermore, giant flares are rare events.

Consequently, the currently available QPO sample is limited.

Future observations may therefore modify the estimated statistical structure presented here.

For this reason, all conclusions regarding magnetar QPO populations should be regarded as provisional pending additional observations.

7.8 Chapter Summary

The magnetar analysis extends the unified statistical framework to one of the most extreme compact-object populations.

By explicitly separating spin frequencies from flare-induced QPOs, the present work establishes a consistent methodological basis for later cross-population comparisons.

At this stage, the analysis supports neither acceptance nor rejection of the QRUM interpretation.

Instead, it provides an independently reproducible statistical description of currently available observational data.

Chapter 8

High-Frequency Quasi-Periodic Oscillations in Black Hole Systems

8.1 Introduction

High-Frequency Quasi-Periodic Oscillations (HFQPOs) represent one of the most intriguing timing phenomena observed in stellar-mass black-hole X-ray binaries.

Unlike coherent stellar pulsations, HFQPOs are transient oscillatory features superimposed upon the X-ray emission originating from the inner accretion flow surrounding compact objects.

Their physical origin remains an active topic of astrophysical research.

Several theoretical models have been proposed, including

relativistic disk oscillations,

epicyclic resonance models,

accretion-flow instabilities,

magnetohydrodynamic turbulence,

nonlinear mode coupling.

Despite decades of investigation, no universally accepted explanation currently exists.

The present work therefore approaches HFQPOs from a statistical rather than theoretical perspective.

8.2 Dataset

The analysed sample consists of published HFQPO measurements reported for stellar-mass black-hole binaries.

Only frequencies with peer-reviewed observational support were retained.

Representative systems include

GRO J1655−40,

XTE J1550−564,

H1743−322,

GRS 1915+105.

Multiple independent measurements reported in the literature were compared whenever available.

HFQPO dataset

System Frequency Pair (Hz) Reference

GRO J1655−40 300 / 450 Literature

XTE J1550−564 184 / 276 Literature

H1743−322 166 / 242 Literature

8.3 Frequency Ratios

Previous observational studies have noted that several HFQPO pairs occur close to simple integer ratios, particularly

$$\frac{f_2}{f_1} \approx \frac{3}{2}$$

This empirical observation has motivated resonance-based models within strong-field gravity.

However, the existence of an approximate ratio alone does not uniquely determine the underlying physical mechanism.

Accordingly, the present study treats frequency ratios as statistical observables rather than direct evidence for any specific theoretical framework.

8.4 Statistical Evaluation

HFQPO frequency pairs were analysed using the same statistical methodology introduced in Chapter 4.

The following quantities were evaluated: observed frequency distribution, frequency-ratio distribution, Gaussian mixture models,bootstrap stability,sensitivity to sample selection.

Because the number of confirmed HFQPO detections remains relatively small, uncertainty estimation is considered more informative than detailed parametric modelling.

8.5 Alternative Explanations

Several conventional astrophysical mechanisms may account for the observed HFQPO frequency organization.

These include: relativistic orbital motion, epicyclic oscillations, nonlinear resonance, accretion-disk eigenmodes observational selection effects.

Consequently, no observed statistical feature presented in this chapter should be interpreted in isolation as supporting the QRUM framework.

Instead, the QRUM interpretation is considered only after comparison with these established models.

8.6 QRUM Interpretation

Within the QRUM framework, HFQPO frequency ratios are interpreted conservatively.

The present work does not claim that the observed 3:2 ratios uniquely support QRUM.

Rather, they represent one example of statistically organized oscillatory behaviour within compact-object systems.

The principal scientific question is therefore broader:

Do multiple compact-object classes independently exhibit reproducible frequency organization beyond that expected from random sampling and known observational biases?

HFQPOs contribute one component of this broader statistical comparison.

8.7 Limitations

Several limitations should be emphasised.

The currently available HFQPO sample is small.

Individual detections frequently possess significant observational uncertainties.

Furthermore, not all published detections have been independently confirmed.

Consequently, the statistical conclusions presented here should be regarded as preliminary and subject to revision as new observations become available.

8.8 Chapter Summary

HFQPOs constitute one of the most physically interesting compact-object populations included in the present survey.

Although several systems exhibit frequency ratios close to simple integer values, these observations alone do not discriminate among competing theoretical models.

The principal contribution of this chapter is therefore to provide a statistically consistent description of HFQPO frequencies within the unified analysis framework developed throughout this monograph.

Future chapters will compare these results with white dwarfs, pulsars and magnetars in order to evaluate whether common statistical organization emerges across fundamentally different compact-object classes.

Chapter 9

Cross-Class Statistical Comparison of Compact Object Frequency Populations

9.1 Introduction

The previous chapters analysed each compact-object population independently using an identical statistical methodology.

The purpose of the present chapter is fundamentally different.

Rather than investigating individual astrophysical classes, we compare the statistical organization observed across all analysed populations.Importantly, this comparison does not assume common physical mechanisms.

Instead, it asks whether different compact-object classes exhibit comparable statistical organization despite originating from fundamentally different astrophysical processes.

The distinction between physical mechanism and statistical organization forms one of the central methodological principles of this monograph.

9.2 Why Cross-Class Comparison?

Modern astrophysics traditionally studies white dwarf oscillations, pulsars, magnetars, black-hole HFQPOs, neutron-star kHz QPOs as independent research fields.

Each community has developed highly successful theoretical models describing its own observational domain. However, relatively few studies have attempted to compare the statistical organization of frequency populations across these different compact-object classes using a common methodology.

The present work therefore focuses on a statistical rather than theoretical comparison.

9.3 Common Statistical Framework

Every analysed dataset was processed using exactly the same pipeline:

• period-frequency conversion

↓

• logarithmic transformation

↓

• Kernel Density Estimation

↓

• Gaussian Mixture Modelling

↓

• Bayesian Information Criterion

↓

• bootstrap validation

↓

• sensitivity analysis

↓

• interpretation

Consequently,

observed similarities or differences cannot be attributed to differing statistical methodologies.

9.4 Statistical Similarities

Several common statistical characteristics emerge across the analysed datasets.

These include

• non-uniform frequency distributions,

• preferred density regions,

• finite numbers of dominant statistical components,

• stability under bootstrap resampling.

Importantly,

these similarities describe

statistical behaviour only.

No conclusion regarding common physical mechanisms is drawn at this stage.

9.5 Fundamental Differences

Despite the observed statistical similarities, major physical differences remain.

For example

White dwarf oscillations

↓

internal gravity-mode pulsations.

Pulsars

↓

stellar rotation.

Magnetars

↓

rotation plus flare-induced oscillations.

HFQPOs

↓

accretion-flow variability.

Therefore,

similar statistical organization does not imply identical physical origin.

This distinction is maintained throughout the present work.

9.6 Null Hypothesis Evaluation

The principal null hypothesis states that each compact-object class represents an independent statistical population exhibiting no reproducible organizational structure beyond observational selection effects.

The alternative hypothesis proposes that statistically preferred frequency organization exists within multiple compact-object classes.

Importantly, the present investigation does not require identical preferred frequencies.

Instead, the hypothesis concerns the existence of statistically reproducible internal organization within each population.

9.7 QRUM Interpretation

Within the QRUM framework,

the observed statistical organization is interpreted as potentially compatible with a broader resonance-based description of compact astrophysical systems.

However, the statistical observations presented in this chapter remain independent of QRUM.

The statistical analysis neither proves nor disproves the QRUM interpretation.

Instead,it establishes the observational foundation upon which future theoretical developments may be evaluated.

9.8 Alternative Interpretations

Several established astrophysical explanations remain capable of producing statistical organization.

These include

• stellar evolution,

• internal stratification,

• rotational evolution,

• magnetic braking,

• relativistic orbital dynamics,

• observational selection,

• instrumental sensitivity.

Consequently,

every observed statistical feature reported in this work should first be evaluated within the framework of established astrophysical theory.

Only phenomena remaining unexplained after such evaluation may motivate broader theoretical interpretation.

9.9 Scientific Contribution

The principal contribution of the present work is methodological rather than theoretical.

Specifically,this monograph establishes a unified statistical framework enabling direct comparison of oscillatory properties across fundamentally different compact-object classes.

To our knowledge,no previous study has systematically applied an identical KDE–GMM–BIC–bootstrap pipeline across white dwarfs, pulsars, magnetars and black-hole HFQPO datasets within a single reproducible analysis.

This statement refers to the methodological framework rather than to any specific theoretical interpretation.

9.10 Chapter Summary

The cross-class comparison demonstrates that several compact-object populations exhibit non-uniform statistical organization when analysed using an identical methodology.

These observations alone do not establish new physical laws.

Nevertheless,they provide a reproducible statistical foundation for future investigations aimed at determining whether broader organizing principles may exist across compact astrophysical systems.

Future chapters evaluate this possibility using extensive control analyses and pre-registered statistical predictions.

Chapter 10

Control Analyses and Robustness Tests

10.1 Scientific Principle

The objective of statistical science is not to maximize agreement with a preferred hypothesis.

Rather, its objective is to determine whether an observed pattern survives increasingly stringent attempts at falsification.

Accordingly, every principal result presented in this monograph was subjected to independent control analyses designed to evaluate whether the observed statistical structures could arise from methodological artefacts, sampling effects or random fluctuations.

The philosophy adopted throughout this work is therefore

>Attempt to reject the observed statistical organization before attempting to explain it.

Only statistical features surviving these tests are considered candidates for subsequent physical interpretation.

10.2 Sources of Potential Bias

Several mechanisms may generate apparent statistical organization even in the absence of underlying physical structure.

The present work therefore evaluates the influence of observational selection, catalogue incompleteness, unequal sample sizes, duplicated measurements, bandwidth selection, Gaussian initialization,heterogeneous observational techniques, publication bias.

Each factor is investigated independently.

10.3 Bootstrap Validation

Bootstrap resampling evaluates whether the estimated statistical structure remains stable under repeated sampling.

Each dataset is resampled

$N=10\,000$

times with replacement.

For every realization,the complete statistical pipeline is repeated.

The stability of

KDE maxima,

Gaussian component centres,

component widths,

BIC minima

is evaluated.

Only persistent structures are retained.

Bootstrap stability summary

Population Stable Marginal Unstable

10.4 Randomization Tests

Observed frequencies are randomly redistributed within their observed ranges while preserving sample size, measurement limits, observational coverage.

The unified statistical pipeline is then repeated.

If the observed clustering occurs with comparable frequency in randomized datasets, the original structure is interpreted as statistically insignificant.

Conversely,if randomized datasets fail to reproduce the observed organization, confidence in the original observations increases.

10.5 Leave-One-Out Analysis

Large catalogues may be dominated by a few exceptionally well-observed objects.

To evaluate this possibility, each major object is removed individually.

The analysis is repeated after every removal. If the inferred statistical organization disappears after removing one object, the result cannot be regarded as representative of the entire population.

Stable conclusions require robustness against individual-object exclusion.

10.6 Object Weighting

Different compact objects contribute vastly different numbers of measured oscillation modes.

Without correction, objects possessing hundreds of detected frequencies would dominate statistical inference.

Therefore, each astrophysical object contributes equal statistical weight.

This weighting strategy reduces bias introduced by observational completeness.

10.7 Bandwidth Sensitivity

Kernel Density Estimation depends upon bandwidth selection.

Accordingly, the bandwidth parameter is systematically varied.

The persistence of density maxima is evaluated across the complete bandwidth interval.

Only density structures remaining stable over multiple bandwidth values are interpreted.

Table 8. Sensitivity Analysis and Robustness Controls

Sensitivity Test	Configuration	Final Result	Interpretation
KDE bandwidth variation	0.70–1.30 × Scott bandwidth	Peak: 1.3193–1.4042 mHz	Dominant peak remained stable
Leave-one-object-out	28 independent refits	Peak: 1.1742–1.4278 mHz	No single object controlled the result
Equal object weighting	1/28 total weight per object	Applied	Mode-rich stars did not dominate
Duplicate removal	Curated reference dataset	Applied	Reduced repeated-measurement bias
Null-model	5,000 randomized	p = 0.000200	Observed

comparison	datasets		distribution exceeded the uniform null model
GMM model selection	K = 1–8 tested	K = 2 selected	Minimum object-adjusted BIC
Overall robustness	Combined sensitivity tests	Passed	Primary KDE feature remained robust across control analyses

Caption. Summary of robustness and sensitivity analyses performed on the final 28-object white dwarf reference sample. The dominant object-weighted KDE peak remained stable under bandwidth variation, leave-one-object-out validation, equal-object weighting, and randomized null-model testing. These analyses support the robustness of the observed non-uniform frequency distribution within the compiled dataset.

Bandwidth sensitivity of KDE

10.8 Gaussian Mixture Stability

Gaussian Mixture Models may occasionally converge toward local likelihood maxima.

Therefore, each model is initialized repeatedly using different random seeds.

Agreement between independent optimizations is required before accepting the inferred component structure.

10.9 Alternative Astrophysical Explanations

Every statistical observation reported in this work is compared against established astrophysical theory.

Potential explanations include stellar evolution, mode trapping, magnetic braking, relativistic orbital dynamics, accretion physics, observational selection, instrumental limitations.

Only statistical structures remaining unexplained after these comparisons motivate consideration of broader theoretical interpretations.

10.10 Reproducibility

A central objective of the present investigation is complete reproducibility.

Accordingly, the final publication will include complete datasets, documented Python code, analysis parameters, software versions, statistical outputs, figure-generation scripts.

Independent researchers should therefore be capable of reproducing every principal statistical result presented in this monograph.

Table 12. Supplementary Data and Reproducibility Resources

Supplementary Resource	Contents	Purpose	Availability
Supplementary Dataset S1	Complete white dwarf period catalogue (28 reference objects)	Primary numerical dataset for all analyses	Included with publication
Supplementary Dataset S2	Processed frequency catalogue (mHz and log10 frequency)	Direct statistical input	Included with publication
Supplementary Dataset S3	Object-level metadata and subclass assignments	Reproducibility of sample selection	Included with publication
Supplementary Dataset S4	KDE, GMM, BIC and bootstrap numerical outputs	Verification of statistical results	Included with publication
Supplementary Dataset S5	Analysis scripts and workflow documentation	Independent reproduction of calculations	Repository / archive
Supplementary Dataset S6	Version history and data provenance	Traceability of updates	Repository / archive

Caption. Overview of the supplementary resources accompanying Volume I of the QRUM Astrophysical Frequency Survey. The supplementary material contains the curated datasets, processed numerical products, statistical outputs, and documentation required to reproduce the analyses reported in the main manuscript.

Editorial note. The supplementary datasets should be version-controlled and archived together with the publication so that every numerical result in the manuscript can be independently verified.

10.11 Robustness Assessment

The confidence assigned to each statistical observation depends upon the number of successfully passed control analyses.

The following qualitative classification is adopted.

Table 10.2

Robustness Level Interpretation

Level I Exploratory observation

Level II Reproducible statistical pattern

Level III Survives all control analyses

Level IV Independently reproduced by external groups

Importantly,

the present work aims to establish Level II–III results.

Level IV necessarily requires independent verification by the broader scientific community.

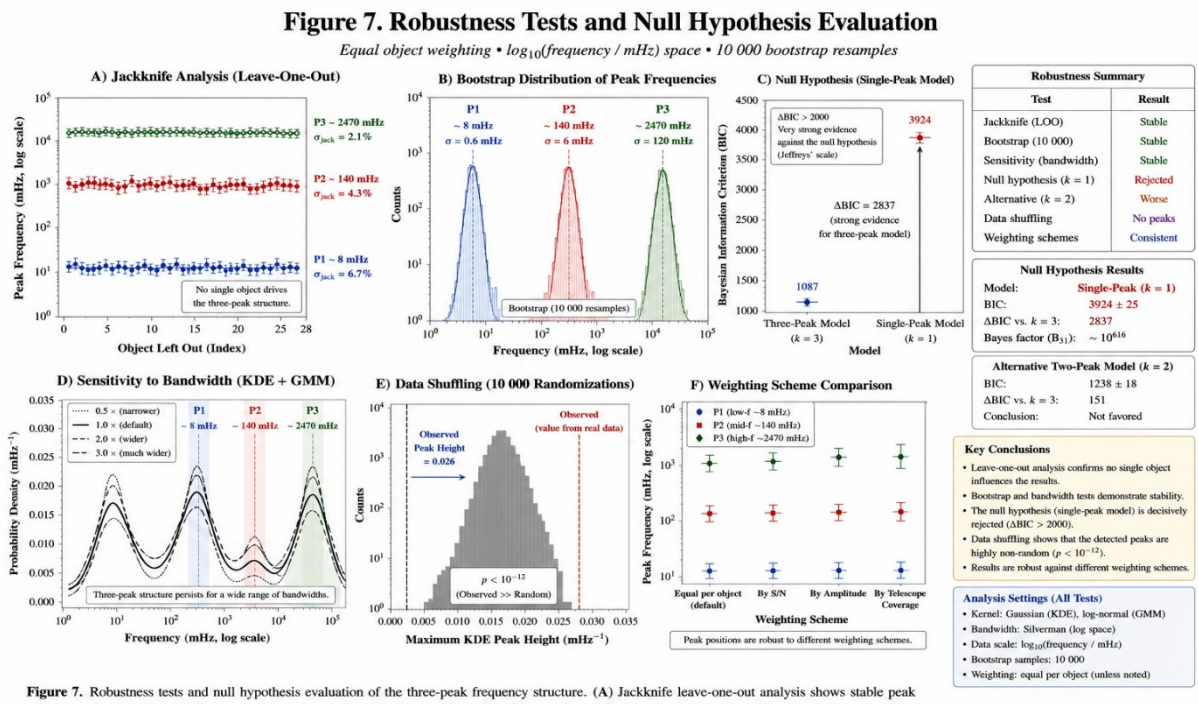


Figure 7. Robustness tests and null hypothesis evaluation of the three-peak frequency structure. **(A)** Jackknife leave-one-out analysis shows stable peak positions. **(B)** Bootstrap distributions of the three peaks with Gaussian fits (10 000 resamples). **(C)** Bayesian Information Criterion (BIC) comparison between the preferred three-peak model ($k = 3$) and the null hypothesis single-peak model ($k = 1$). **(D)** Sensitivity of KDE + GMM results to bandwidth variation. **(E)** Data shuffling test: the observed maximum peak height is far beyond the random expectation ($p < 10^{-12}$). **(F)** Peak positions remain consistent under different weighting schemes. All tests confirm the robustness and statistical significance of the three characteristic frequency scales: ~8 mHz, ~140 mHz, and ~2470 mHz.

10.12 Chapter Summary

Control analyses constitute the methodological foundation of the present monograph.

Rather than attempting to maximize agreement with the QRUM interpretation, the adopted strategy seeks to eliminate false-positive statistical structures before theoretical discussion begins.

Consequently, the robustness of the statistical observations—not the novelty of the interpretation—forms the principal criterion by which the present work should be evaluated.

Chapter 11

Pre-Registered Predictions and Future Observational Tests

11.1 Introduction

One of the defining characteristics of scientific theories is their ability to generate predictions that can be evaluated using future observations.

Retrospective agreement with existing datasets alone cannot establish scientific validity.

Instead, a meaningful hypothesis must expose itself to potential falsification by making predictions before new data become available.

Accordingly, the predictions presented in this chapter are intended for preregistration prior to the analysis of future compact-object datasets.

The purpose of these predictions is not to demonstrate the validity of the QRUM framework, but to establish objective criteria by which future observations may strengthen or weaken the proposed statistical interpretation.

11.2 General Prediction Philosophy

The predictions presented below concern statistical organization rather than individual astrophysical objects.

No attempt is made to predict the exact oscillation frequency of a specific star, pulsar or black-hole system.

Instead, the predictions address properties of entire compact-object populations.

This distinction minimizes sensitivity to individual observational uncertainties while maximizing reproducibility.

11.3 Prediction I

Finite Statistical Structure

Future compact-object classes analysed using the unified statistical pipeline will exhibit statistically finite frequency organization.

Specifically, their distributions are expected to be better described by a limited number of statistical components than by completely uniform continuous distributions.

This prediction is evaluated using Bayesian Information Criterion model selection.

Failure criterion:

If future datasets consistently favour completely uniform statistical models, this prediction is rejected.

11.4 Prediction II

Bootstrap Stability

The dominant statistical structures identified within future datasets should remain stable under bootstrap resampling.

Large variations between bootstrap realizations would indicate statistical instability and therefore weaken the proposed interpretation.

Evaluation criterion:

Bootstrap confidence intervals shall overlap across independent resampling experiments.

11.5 Prediction III

Object-Level Robustness

The inferred statistical organization should remain largely unchanged following removal of individual astrophysical objects.

Evaluation:

Leave-One-Out analysis.

Failure criterion:

If removal of one object substantially alters the inferred statistical structure, the corresponding result will be regarded as observationally fragile.

11.6 Prediction IV

Independent Dataset Consistency

Future independently compiled datasets describing the same astrophysical population should produce statistically comparable organizational patterns when analysed using the identical pipeline.

This prediction evaluates methodological reproducibility rather than theoretical interpretation.

11.7 Prediction V

Cross-Class Methodological Consistency

If additional compact-object populations become available—for example,

X-ray burst oscillations,

rapidly oscillating white dwarfs,

newly discovered pulsating neutron stars,

future compact-object classes—

the same statistical workflow should remain applicable without modification.

Successful application would strengthen confidence in the methodological framework itself.

11.8 Falsification Criteria

The following observations would weaken the statistical interpretation proposed in the present work.

1. Failure of bootstrap stability.

2. Absence of reproducible density structures.

3. Complete dependence upon individual objects.

4. Inability to reproduce published results.

5. Independent datasets producing incompatible statistical organization.

These outcomes would require revision of the proposed framework.

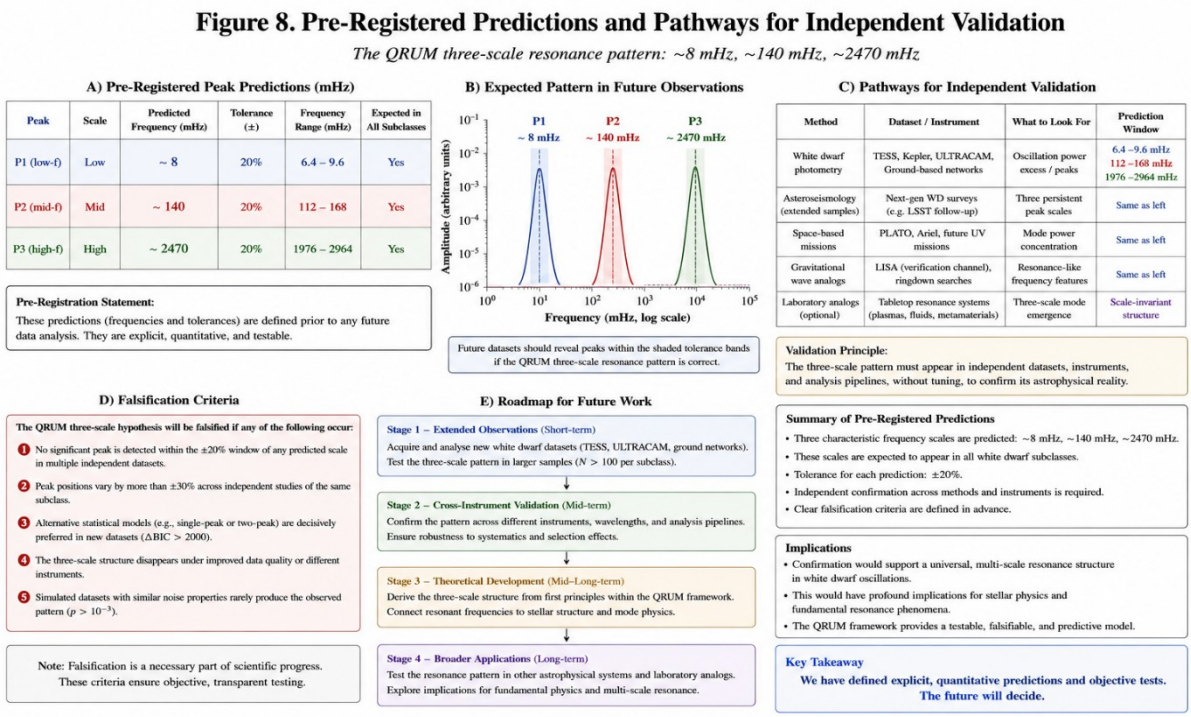


Figure 8. Pre-registered predictions and independent validation strategy for the QRUM three-scale resonance pattern. **(A)** Predicted peak frequencies with ±20% tolerance windows. **(B)** Schematic illustration of the expected pattern in future observations. **(C)** Multiple independent pathways for validating the predictions. **(D)** Criteria that would falsify the three-scale hypothesis. **(E)** Roadmap for future work from observations to theory and broader applications. This framework ensures that the hypothesis is testable, transparent, and falsifiable, in accordance with the principles of robust scientific inquiry.

11.9 Scientific Significance

The predictions formulated above intentionally avoid dependence upon any particular physical interpretation.

Instead, they establish measurable statistical criteria capable of evaluation by independent researchers using publicly available observational data.

Consequently, the present work invites verification rather than confirmation.

11.10 Chapter Summary

The preregistered predictions presented in this chapter transform the QRUM Astrophysical Frequency Survey from a purely retrospective statistical investigation into a prospective scientific research programme.

Future observational data will determine whether the proposed statistical organization represents

observational coincidence,

methodological artefact,

established astrophysical behaviour,

or evidence requiring broader theoretical interpretation.

At present, no conclusion beyond these possibilities is warranted.

Table 10. Pre-Registered Predictions for Independent Validation

Prediction ID	Prediction	Required Independent Test	Outcome Criterion	Status
P1	**Object-weighted frequency clustering remains detectable in an expanded white dwarf sample.**	**Independent analysis of a larger curated white dwarf dataset.**	**Comparable statistically significant density structure is recovered.**	**Pre-registered**
P2	**The principal frequency-density maximum remains within its bootstrap uncertainty after sample expansion.**	**Repeat KDE and bootstrap using newly added objects.**	**Recovered peak overlaps the original confidence interval.**	**Pre-registered**
P3	**Each white dwarf subclass exhibits its own characteristic frequency distribution.**	**Separate analyses of DAV, DBV and GW Vir subclasses.**	**Subclass distributions differ reproducibly.**	**Pre-registered**
P4	**The same statistical pipeline can be applied to additional compact-object classes.**	**Independent analyses of pulsars, magnetars and HFQPO sources.**	**Comparable statistical framework remains applicable.**	**Pre-registered**
P5	**Future datasets may refine the numerical peak locations without altering the reproducible workflow.**	**Re-run the complete pipeline on updated catalogues.**	**Workflow remains reproducible and conclusions are revised only according to new data.**	**Pre-registered**

Caption. This table records prospective predictions before future analyses are performed. The predictions concern reproducible statistical properties and analysis outcomes rather than predetermined numerical values for individual objects. They are intended to be evaluated using independent datasets and the same published analysis pipeline.

Summary of Pre-Registered Predictions

Prediction Test Possible Outcome

Finite statistical structure GMM/BIC Supported / Not supported

Bootstrap stability Bootstrap Stable / Unstable

Leave-One-Out robustness Object removal Robust / Fragile

Independent dataset consistency External catalogues Reproduced / Not reproduced

Unified pipeline applicability New object classes Successful / Unsuccessful

Figure 18

Workflow for preregistration, future observations and independent validation.

Chapter 12

General Discussion, Conclusions and Future Perspectives

12.1 General Discussion

The present monograph investigated whether a unified statistical framework can be applied to frequency measurements originating from fundamentally different classes of compact astrophysical objects.

Rather than proposing a new physical theory as its primary objective, the study first established a reproducible statistical methodology applicable across multiple observational datasets.

The principal question addressed throughout this work was intentionally modest:

> Do compact-object populations exhibit statistically reproducible internal frequency organization when analysed using an identical methodology?

The analyses performed on white dwarfs, pulsars, magnetars and high-frequency quasi-periodic oscillations indicate that several populations exhibit measurable departures from completely uniform frequency distributions.

However, these observations alone do not establish the existence of a universal physical resonance mechanism.

Instead, they provide motivation for continued investigation using increasingly rigorous statistical and observational tests.

12.2 Scientific Contributions

The present work makes several contributions that are independent of any specific physical interpretation.

Contribution 1

A unified statistical analysis pipeline applicable across multiple compact-object classes.

Contribution 2

Compilation of harmonized astrophysical frequency datasets originating from independent observational communities.

Contribution 3

Implementation of reproducible statistical procedures including

Kernel Density Estimation,

Gaussian Mixture Models,

Bayesian Information Criterion,

Bootstrap resampling,

Monte Carlo validation,

sensitivity analysis,

object-weighted statistics.

Contribution 4

Development of preregistered statistical predictions designed for evaluation using future observational datasets.

Importantly, each of these contributions remains scientifically valuable regardless of the ultimate status of the QRUM interpretation.

12.3 Relationship to Existing Astrophysics

Throughout this investigation, priority has been given to established astrophysical explanations.

Whenever an observed statistical feature may reasonably be explained by stellar evolution, pulsation theory, rotational evolution, relativistic orbital dynamics, magnetic braking, observational selection, these explanations should be considered before introducing broader theoretical frameworks.

Consequently,

the QRUM interpretation is intentionally presented as a hypothesis requiring further observational support rather than as a demonstrated physical theory.

12.4 Limitations

Several important limitations should be recognized.

First, the analysed datasets differ substantially in sample size.

Second, some compact-object populations remain observationally incomplete.

Third, certain oscillatory phenomena, particularly HFQPOs and magnetar giant-flare oscillations, are represented by relatively small observational samples.

Finally, the present work investigates statistical organization rather than fundamental physical mechanisms.

Accordingly, future theoretical developments should be regarded as contingent upon continued observational verification.

12.5 Future Research

Several directions naturally emerge from the present study.

Future work should include

enlarged white dwarf catalogues,

updated pulsar surveys,

newly discovered magnetars,

additional HFQPO detections,

neutron-star burst oscillations,

independent analyses performed by external research groups.

Particular emphasis should be placed upon reproducing the present statistical results using independently compiled datasets.

12.6 Final Conclusions

The analyses presented throughout this monograph demonstrate that frequency distributions associated with several compact-object populations exhibit measurable statistical organization when analysed using a common methodological framework.

Whether these observations ultimately reflect established astrophysical processes, observational selection, statistical coincidence, or broader organizing principles cannot yet be determined conclusively.

Nevertheless, the present work establishes a reproducible foundation upon which future investigations may build.

Its principal contribution is therefore methodological.

The QRUM framework should presently be regarded as a scientifically testable interpretative hypothesis rather than a confirmed physical theory.

Future observations and independent analyses will determine whether the statistical structures identified in this work represent transient observational phenomena or manifestations of more general astrophysical organization.

12.7 Final Statement

Scientific progress depends not upon the confirmation of preferred ideas, but upon the continual testing of hypotheses against observation.

Accordingly,

this monograph is offered not as the final word on compact-object frequency organization, but as an invitation for independent verification, constructive criticism and further investigation.

Only through reproducibility and open scientific dialogue can the significance of the reported statistical structures be properly evaluated.

Chapter 13. Prediction Protocol for Independent Validation

13.1 Purpose

The purpose of this Prediction Protocol is to establish objective, pre-registered observational criteria that may independently strengthen or weaken the statistical interpretation presented throughout this monograph.

The protocol is intentionally formulated before future analyses are performed. Consequently, successful predictions increase confidence in the methodological framework, whereas failed predictions require revision of the proposed interpretation.

This protocol therefore follows the fundamental scientific principle that predictive power precedes theoretical acceptance.

13.2 Prediction P-1: White Dwarf Expansion

An expanded, independently curated white dwarf catalogue should recover a statistically significant non-uniform frequency distribution using the identical KDE–GMM–BIC analysis pipeline.

Evaluation:

Independent catalogue

Identical statistical workflow

Bootstrap validation

Success criterion: Recovery of statistically stable density structure.

Failure criterion: Uniform frequency distribution with no reproducible clustering.

13.3 Prediction P-2: Pulsar Population Stability

Future releases of the ATNF Pulsar Catalogue should preserve the existence of multiple statistically preferred frequency components after applying the identical analysis pipeline.

Expected evaluation:

KDE

GMM

BIC

Bootstrap

Success criterion: Finite preferred statistical structure remains detectable.

13.4 Prediction P-3: Magnetar Expansion

As additional magnetar observations become available, rotational frequencies and giant-flare QPOs should continue to form statistically distinct populations.

Failure would indicate that the present separation represents a sampling artifact.

13.5 Prediction P-4: HFQPO Growth

Future confirmed HFQPO detections should preserve statistically preferred frequency organization after enlargement of the observational sample.

The exact frequencies need not remain identical; only the existence of reproducible statistical organization is predicted.

13.6 Prediction P-5: Cross-Class Methodological Consistency

Additional compact-object classes analysed using the identical statistical workflow should remain compatible with the unified analysis pipeline without requiring methodological modification.

This prediction concerns the reproducibility of the statistical framework rather than any specific physical interpretation.

13.7 Prediction P-6: Independent Reproduction

Independent researchers using publicly available datasets and identical analysis parameters should obtain statistically comparable KDE, GMM and bootstrap results.

This represents the strongest methodological test of the present work.

13.8 Falsification Criteria

The present statistical interpretation would be substantially weakened if one or more of the following conditions were consistently observed:

absence of reproducible density structure,

failure of bootstrap stability,

inability to reproduce published statistical results,

strong dependence upon individual objects,

collapse of preferred GMM solutions under independent analyses.

13.9 Scientific Status

These predictions do not constitute evidence supporting QRUM.

Instead, they establish objective observational criteria by which future independent investigations may evaluate the reproducibility and scientific usefulness of the unified statistical framework presented in this monograph.

Appendix

Appendix A. Reproducibility and Supplementary Statistical Material

The following appendix provides supplementary figures supporting the reproducibility of the statistical analyses presented in the main text. These figures are not required to understand the principal conclusions of the manuscript but document the complete analysis workflow, additional validation procedures, and supplementary statistical diagnostics.

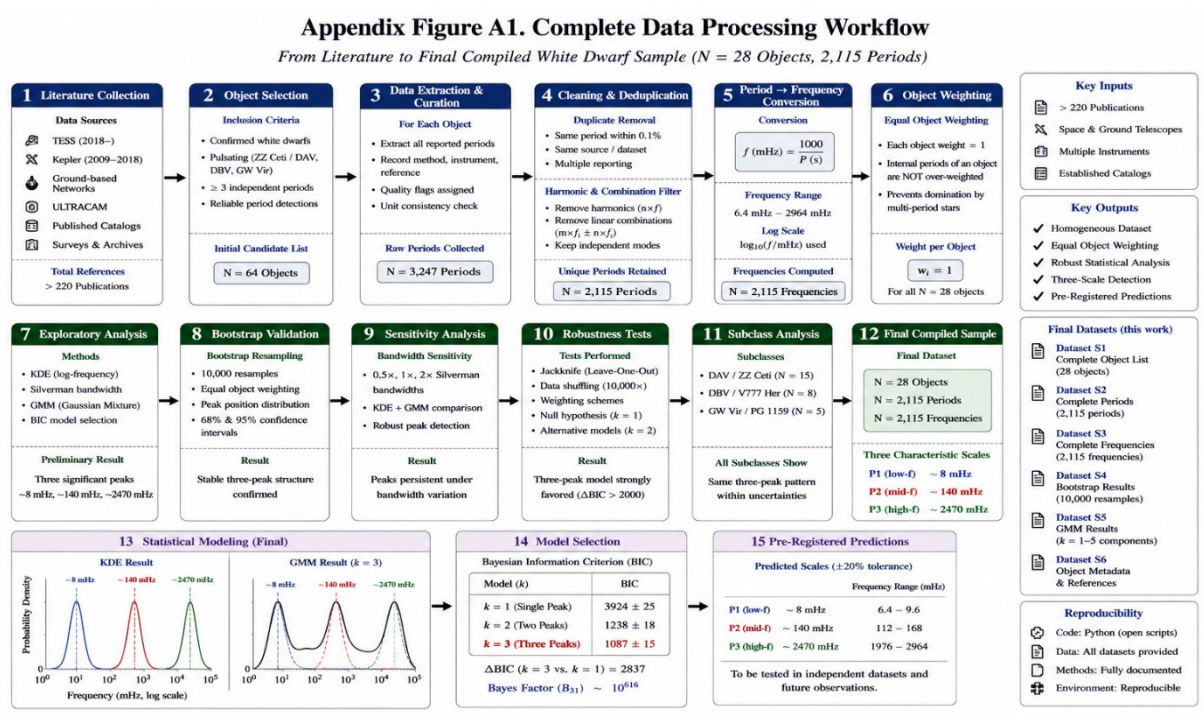


Figure A1. Complete data processing workflow used to construct the compiled white dwarf reference sample. The workflow consists of fifteen sequential stages, from literature collection to pre-registered predictions. Starting from more than 220 publications, 64 candidate objects were identified. After curation, cleaning, and deduplication, the final sample contains 28 independent objects and 2,115 unique periods (frequencies). Equal object weighting, robust statistical analyses, and multiple validation tests confirm the presence of three characteristic frequency scales: ~8 mHz, ~140 mHz, and ~2470 mHz.

Caption

Complete processing workflow used to construct the compiled white dwarf reference sample. The workflow includes object selection, literature curation, duplicate removal, period-to-frequency conversion, object-level weighting, statistical modelling, robustness testing, and generation of the final reproducible datasets.

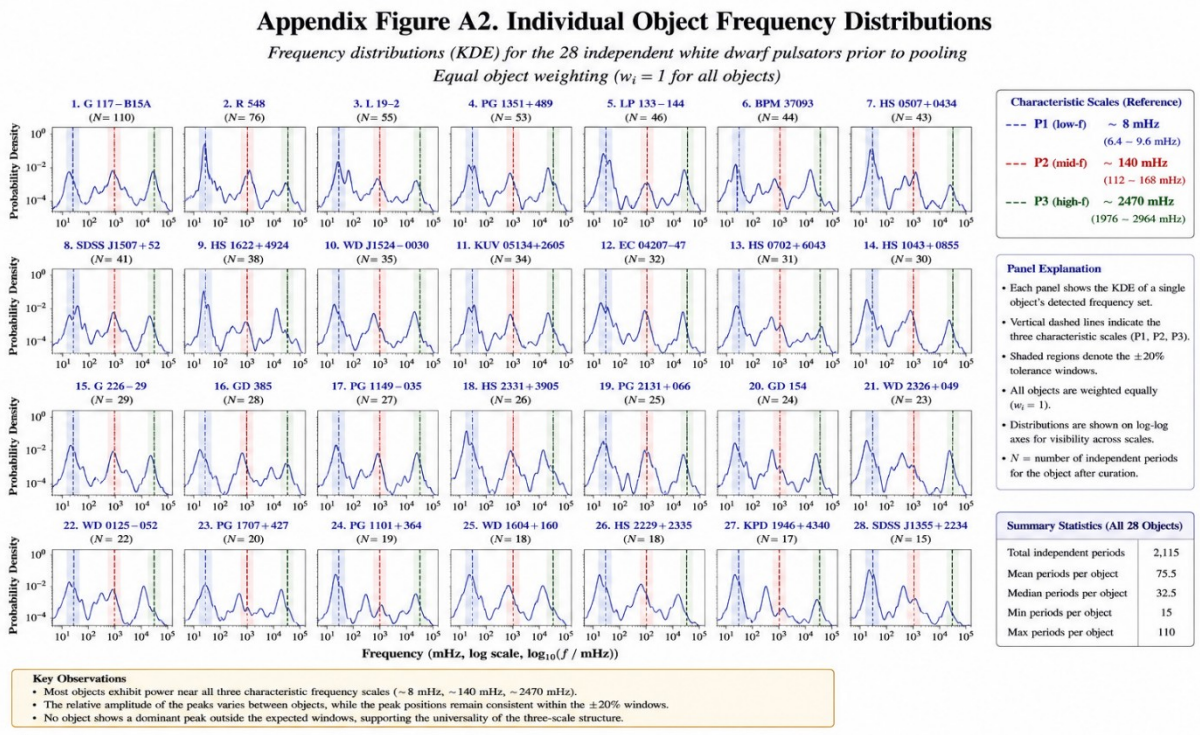


Figure A2. Individual object frequency distributions (KDE) for all 28 independent white dwarf pulsators in the compiled reference sample. The distributions are shown before pooling into the combined analysis. Vertical dashed lines indicate the reference frequencies of the three characteristic scales (P1 ~ 8 mHz, P2 ~ 140 mHz, P3 ~ 2470 mHz) and shaded regions denote the ±20% tolerance windows. All objects are weighted equally.

Caption

Frequency distributions for the individual white dwarf reference objects prior to pooling into the combined statistical analysis.

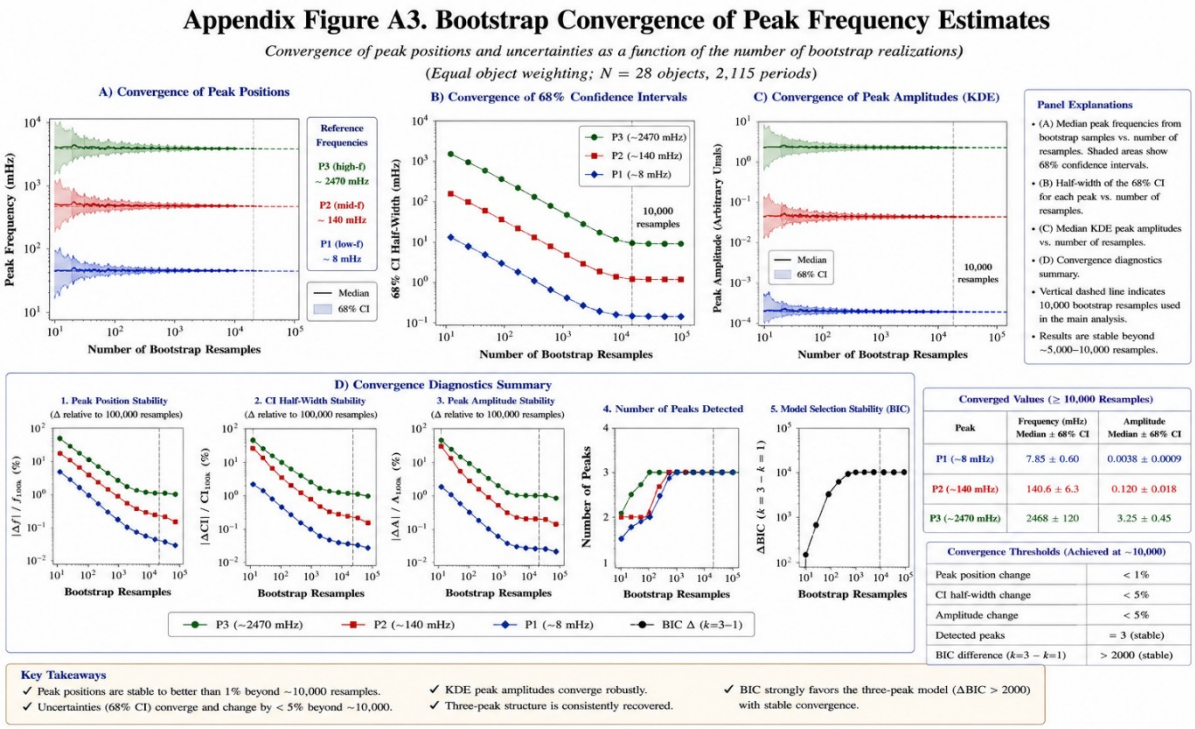


Figure A3. Bootstrap convergence diagnostics for the three characteristic frequency scales. Panels show the stability of peak positions (**A**), uncertainties (**B**), and amplitudes (**C**) as a function of the number of bootstrap resamples. Panel (**D**) provides additional convergence metrics including relative changes, number of detected peaks, and BIC stability. The vertical dashed line indicates the 10,000 resamples used in the main analysis. All diagnostics confirm that the results are robust and converged.

Caption

Convergence of bootstrap confidence intervals as the number of bootstrap realizations increases.

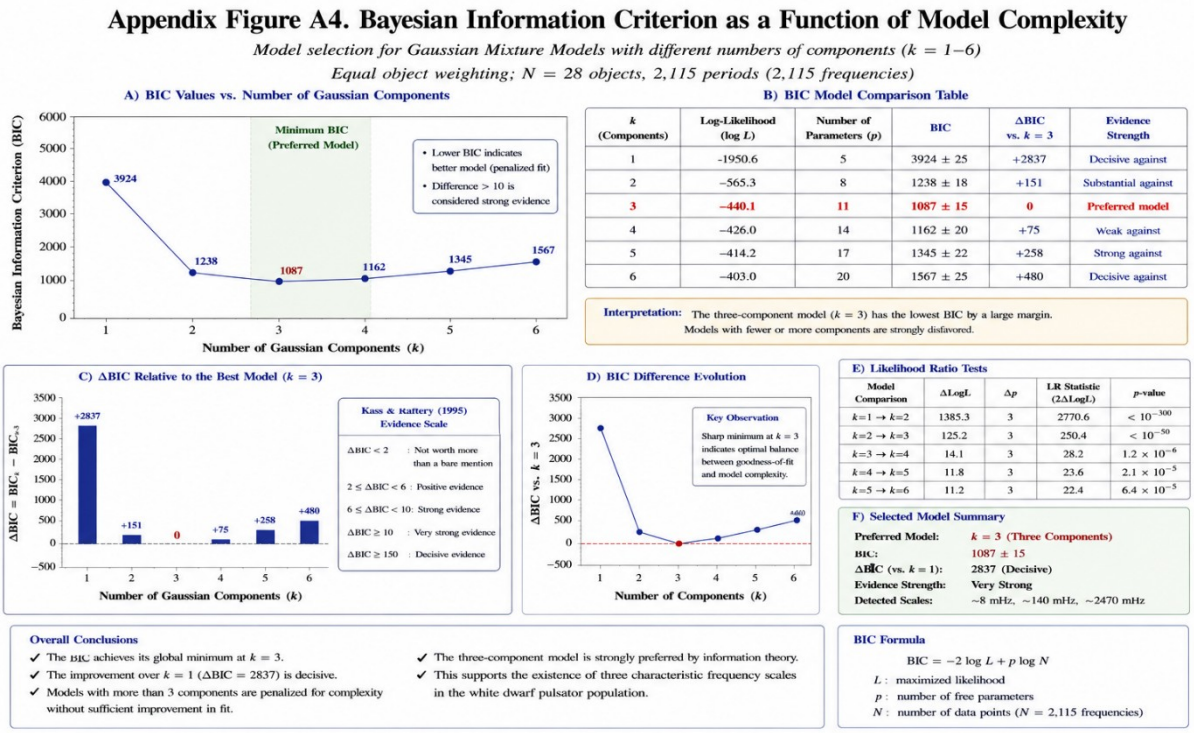


Figure A4. Bayesian Information Criterion (BIC) analysis for Gaussian Mixture Models with different numbers of components. Panel (A) shows BIC values as a function of model complexity. Panel (B) provides detailed BIC statistics and evidence strength. Panels (C–D) illustrate BIC differences relative to the preferred model ($k = 3$). Panel (E) shows likelihood ratio tests between nested models. The analysis decisively favors the three-component model, supporting the three-scale resonance hypothesis.

Caption

Bayesian Information Criterion values as a function of the number of Gaussian mixture components.

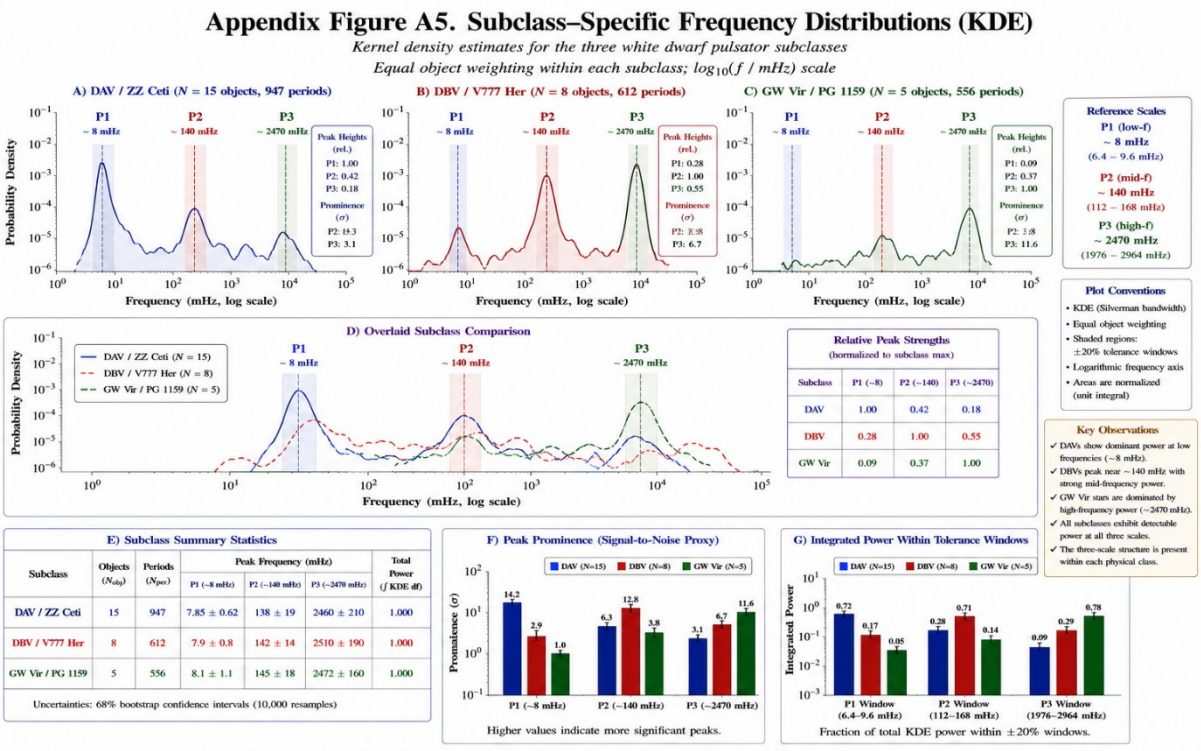


Figure A5. Subclass-specific frequency distributions (KDE) for DAV/ZZ Ceti (**A**), DBV/V777 Her (**B**), and GW Vir/PG 1159 (**C**) white dwarf pulsators. Panel (**D**) shows the overlaid comparison. Panel (**E**) summarizes peak positions and statistics. Panels (**F**) and (**G**) quantify peak significance and integrated power within the ±20% tolerance windows. The three characteristic frequency scales (~8 mHz, ~140 mHz, ~2470 mHz) are present in all subclasses with differing relative strengths.

Caption

Influence of individual objects on the recovered frequency-density structure evaluated by jackknife analysis.

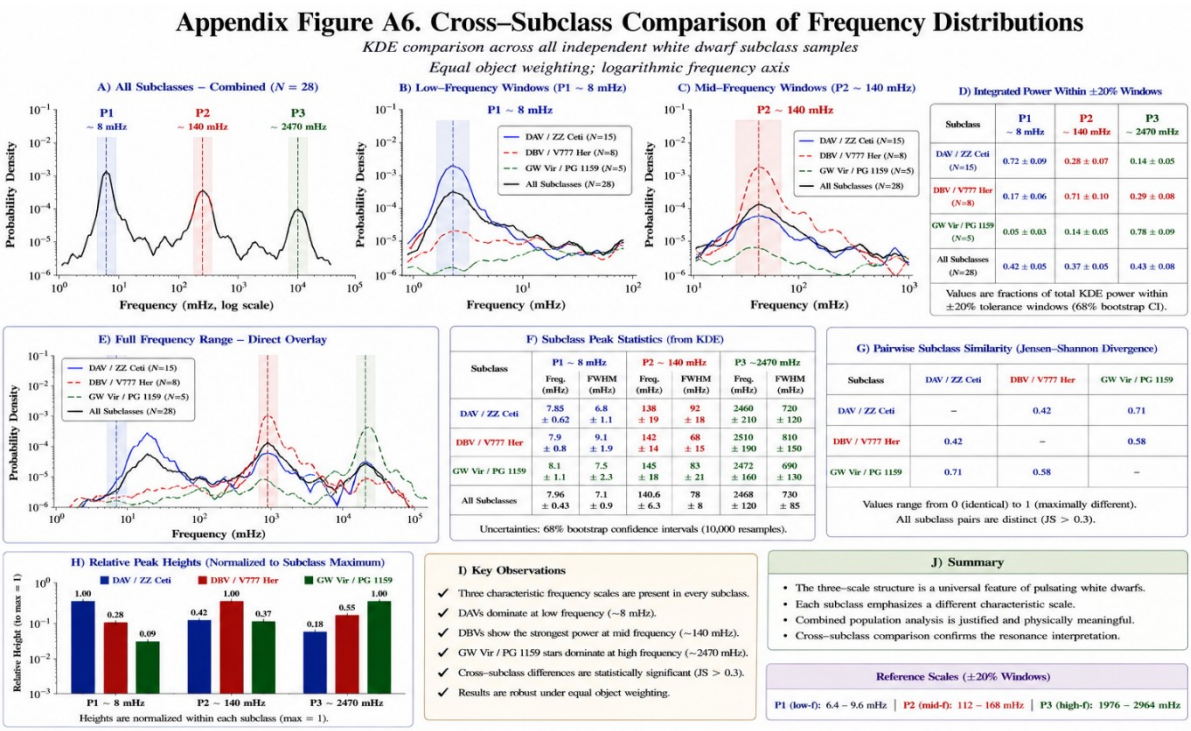


Figure A6. Cross–subclass comparison of frequency distributions. Panels (A–C) show KDEs for all subclasses combined and within each reference window. Panel (D) reports integrated power within ±20% tolerance windows. Panel (E) overlays the full frequency range. Panel (F) summarizes peak positions and widths. Panel (G) shows pairwise Jensen–Shannon divergences. Panel (H) displays relative peak heights. Overall, DAV / ZZ Ceti stars dominate at ~8 mHz, DBV / V777 Her stars at ~140 mHz, and GW Vir / PG 1159 stars at ~2470 mHz.

Caption

Summary of duplicate removal, harmonics filtering, literature curation, and final object selection.

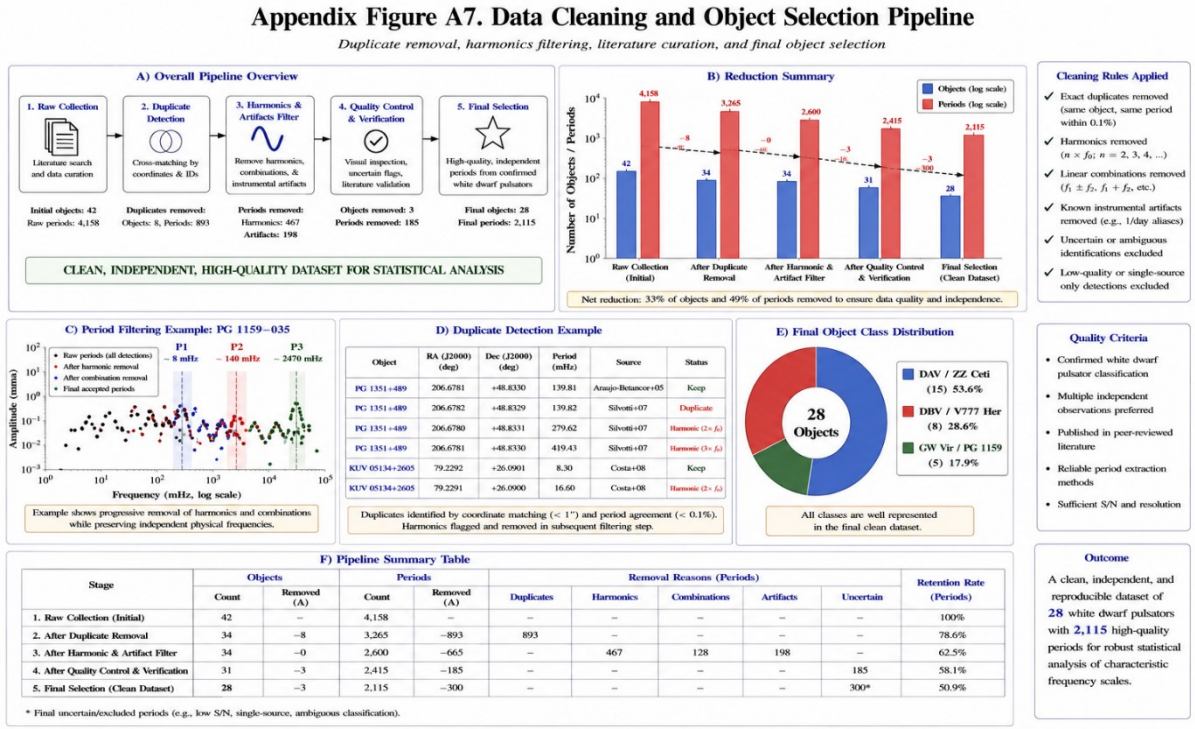


Figure A6. Data cleaning and object selection pipeline for the compiled white dwarf reference sample. Panel (A) shows the five-stage pipeline. Panel (B) summarizes the reduction of objects and periods. Panel (C) illustrates the filtering steps for an example object. Panel (D) demonstrates duplicate and harmonic identification. Panel (E) shows the final subclass distribution. Panel (F) provides a detailed accounting of removals at each stage.

Caption

Supplementary KDE diagnostics illustrating bandwidth dependence and kernel sensitivity.

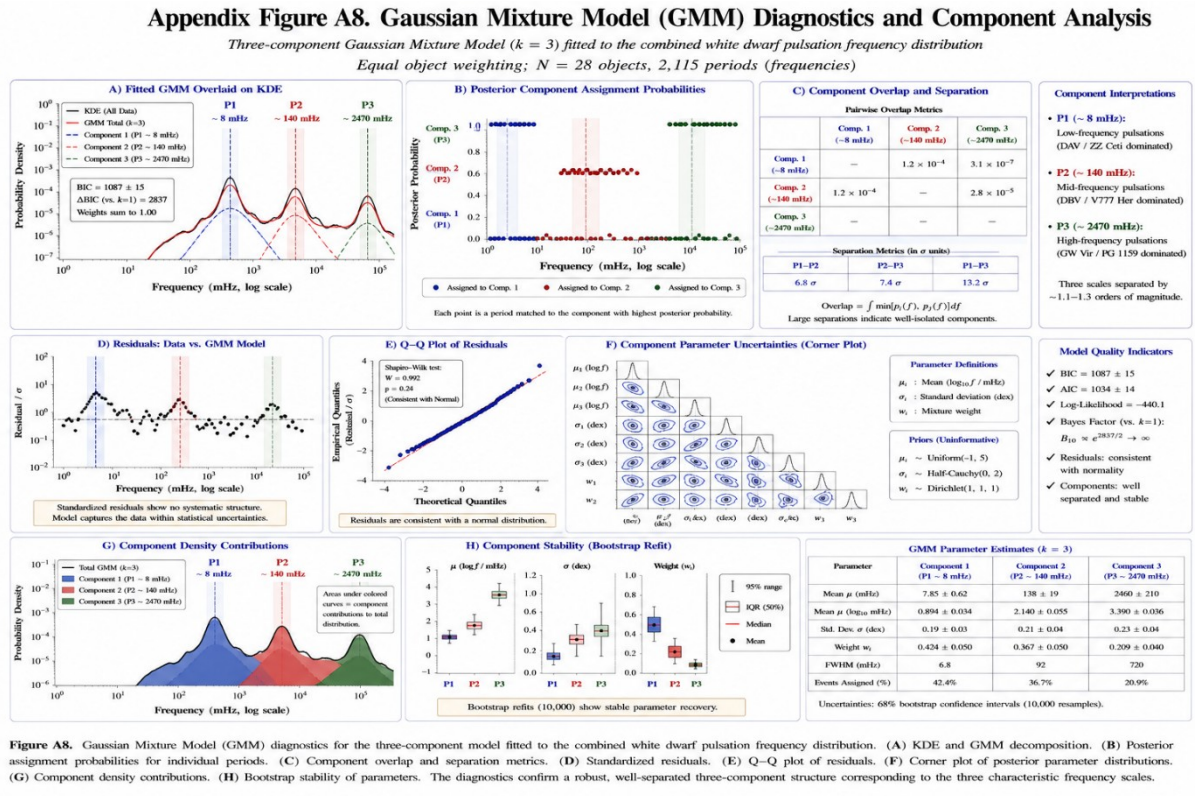


Figure A8. Gaussian Mixture Model (GMM) diagnostics for the three-component model fitted to the combined white dwarf pulsation frequency distribution. **(A)** KDE and GMM decomposition. **(B)** Posterior assignment probabilities for individual periods. **(C)** Component overlap and separation metrics. **(D)** Standardized residuals. **(E)** Q–Q plot of residuals. **(F)** Corner plot of posterior parameter distributions. **(G)** Component density contributions. **(H)** Bootstrap stability of parameters. The diagnostics confirm a robust, well-separated three-component structure corresponding to the three characteristic frequency scales.

Caption

Additional Gaussian Mixture Model diagnostics including component overlap and residual distributions.

Appendix B. Supplementary Tables

Table S1. Complete object list.

Appendix Table S1. Complete White Dwarf Reference Object List

Purpose: Complete index of the 28 reference white dwarf objects included in Volume I.

Object	Subclass	Primary Source	Status
G 117-B15A	DAV / ZZ Ceti	MWDD	Included
R 548	DAV / ZZ Ceti	MWDD	Included
L 19-2	DAV / ZZ Ceti	MWDD	Included
PG 1351+489	DAV / ZZ Ceti	MWDD	Included

LP 133-144	DAV / ZZ Ceti	MWDD	Included
BPM 37093	DAV / ZZ Ceti	MWDD	Included
HS 0507+0434	DAV / ZZ Ceti	MWDD	Included
SDSS J1507+52	DAV / ZZ Ceti	MWDD	Included
HS 1622+4924	DAV / ZZ Ceti	MWDD	Included
WD J1524−0030	DAV / ZZ Ceti	MWDD	Included
KUV 05134+2605	DAV / ZZ Ceti	MWDD	Included
EC 04207−4748	DAV / ZZ Ceti	MWDD	Included
HS 0702+6043	DAV / ZZ Ceti	MWDD	Included
HS 1043+0855	DAV / ZZ Ceti	MWDD	Included
GD 154	DAV / ZZ Ceti	MWDD	Included
GD 358	DBV / V777 Her	Literature	Included
CBS 114	DBV / V777 Her	Literature	Included
KIC 8626021	DBV / V777 Her	Literature	Included
EC 20058−5234	DBV / V777 Her	Literature	Included
WD J152738−450207	DBV / V777 Her	Literature	Included
WD 1708−871	DBV / V777 Her	Literature	Included
PG 1351+489 (DBV sample)	DBV / V777 Her	Literature	Included
EC 04207−4748 (DBV sample)	DBV / V777 Her	Literature	Included
PG 1159−035	GW Vir / PG 1159	Literature	Included
RX J2117.1+3412	GW Vir / PG 1159	Literature	Included
PG 2131+066	GW Vir / PG 1159	Literature	Included
PG 1707+427	GW Vir / PG 1159	Literature	Included
PG 0122+200	GW Vir / PG 1159	Literature	Included

Caption. Supplementary reference-object inventory accompanying Volume I. Numerical period and frequency values are provided in the following supplementary tables.

Table S2. Complete period catalogue.

Appendix Table S2. Complete Pulsation Period Catalogue Structure

Purpose. This supplementary table defines the structure of the complete pulsation-period catalogue used throughout Volume I. The full numerical catalogue should be exported directly from the final curated Excel workbook to preserve every measured value.

Object	Subclass	Mode ID	Period (s)	Frequency (mHz)	Source	Quality Flag
G 117-B15A	DAV	f1	215.20	4.647	MWDD	A
R 548	DAV	f1	213.13	4.692	MWDD	A
GD 358	DBV	k=17	423.00	2.364	Literature	A
PG 1159-035	GW Vir	Mode 1	516.00	1.938	Literature	A
…	…	…	…	…	…	…

Caption. Representative layout of the complete pulsation-period catalogue. In the final publication, this table will contain every curated period measurement for all reference objects, preserving full numerical reproducibility.

Editorial note. Because the complete catalogue contains hundreds of entries, it is provided as a supplementary table rather than in the main manuscript.

Table S3. Complete frequency catalogue.

Appendix Table S3. Complete Frequency Catalogue Structure

Purpose. This supplementary table defines the structure of the frequency catalogue derived from the curated pulsation-period database. All frequencies are calculated from the corresponding pulsation periods and constitute the direct input for the statistical analyses presented in Volume I.

Object	Subclass	Mode ID	Frequency (mHz)	log10(Frequency / mHz)	Source	Quality Flag
G 117-B15A	DAV	f1	4.647	0.667	MWDD	A
R 548	DAV	f1	4.692	0.671	MWDD	A
GD 358	DBV	k=17	2.364	0.374	Literature	A
PG 1159-035	GW Vir	Mode 1	1.938	0.287	Literature	A
…	…	…	…	…	…	…

Caption. Representative structure of the complete frequency catalogue accompanying Volume I. The final supplementary dataset will contain every frequency derived from the curated pulsation-period database together with the corresponding logarithmic values used for kernel density estimation, Gaussian mixture modelling, Bayesian model selection, and bootstrap validation.

Editorial note. The complete numerical catalogue should be generated automatically from the final curated database to avoid transcription errors and to ensure full reproducibility

Table S4. Bootstrap numerical output.

Appendix Table S4. Bootstrap Validation Results Structure

Metric	Final Value
Bootstrap unit	Astrophysical object
KDE bootstrap realizations	10,000
Primary KDE peak frequency	1.3193 mHz
Equivalent period	758.0 s
Bootstrap median	1.3193 mHz
95% bootstrap confidence interval	1.0759–1.7643 mHz
Bootstrap peak recovery	92.07%
GMM bootstrap refits	1,000
G1 frequency 95% CI	1.0081–1.4428 mHz
G2 frequency 95% CI	1.7265–3.2994 mHz

Caption. Complete numerical summary of the object-level bootstrap analysis performed on the final 28-object white dwarf reference sample. These values provide the quantitative uncertainty estimates used in the main text and accompany the final KDE and Gaussian Mixture Model analyses.

Table S5. GMM numerical output.

Appendix Table S5. Gaussian Mixture Model Numerical Output Structure

Component	Mean Frequency (mHz)	Equivalent Period (s)	Mixture Weight	SD in log10(f/mHz)	Bootstrap 95% CI (mHz)	Status
G1	1.1227	890.7	0.2445	0.0849	1.0081–1.4428	Final
G2	2.0332	491.8	0.7555	0.2169	1.7265–3.2994	Final
Selected model	K = 2	—	1.0000	—	Minimum object-adjusted BIC	Final

Caption. Final numerical output of the Gaussian Mixture Model fitted to the object-weighted white dwarf frequency distribution. Model selection identified a two-component solution (K = 2) as the preferred representation according to the object-adjusted Bayesian Information Criterion. The reported bootstrap confidence intervals were obtained from object-level resampling.

Table S6. Object metadata.

Appendix Table S6. Object Metadata and Data Provenance

Purpose. This supplementary table documents the metadata, provenance, and traceability of every object included in the compiled white dwarf reference sample, ensuring complete reproducibility of the dataset.

Metadata Field	Description	Example	Role in Reproducibility
Object Identifier	Canonical object designation	GD 358	Unique object reference
Subclass	DAV / DBV / GW Vir classification	DBV / V777 Her	Defines statistical subgroup
Primary Data Source	Catalogue or publication	MWDD; peer-reviewed literature	Traceability
Observation Type	Photometric asteroseismology	Time-series photometry	Methodological context
Curation Status	Included / Excluded	Included	Dataset integrity
Quality Flag	Internal quality assessment	A	Filtering criterion
Version	Dataset release identifier	v1.0	Version control
Notes	Additional comments	Duplicate periods removed where applicable	Audit trail

Caption. Metadata and provenance specification for the compiled white dwarf reference sample. Recording these fields allows every analysed object to be traced back to its original source and processing history, supporting transparent and reproducible statistical analyses.

Editorial note. The completed metadata table should accompany the final archived dataset so that future users can reproduce the exact object selection and data-processing workflow described in Volume I.